```python
import sys
print(sys.executable)
print(sys.version)
```

```
D:\Python_3.13\python.exe
3.13.13 (tags/v3.13.13:01104ce, Apr  7 2026, 19:25:48) [MSC v.1944 64
bit (AMD64)]
```

```python
import sys
!"{sys.executable}" -m pip install --upgrade pip
!"{sys.executable}" -m pip install pandas
```

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Requirement already satisfied: pip in D:\Python_3.13\Lib\site-packages
(26.0.1)
Collecting pandas
  Downloading pandas-3.0.2-cp313-cp313-win_amd64.whl.metadata (19 kB)
Collecting numpy>=1.26.0 (from pandas)
  Downloading numpy-2.4.4-cp313-cp313-win_amd64.whl.metadata (6.6 kB)
Requirement already satisfied: python-dateutil>=2.8.2 in D:\
Python_3.13\Lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: tzdata in D:\Python_3.13\Lib\site-
packages (from pandas) (2026.1)
Requirement already satisfied: six>=1.5 in D:\Python_3.13\Lib\site-
packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Downloading pandas-3.0.2-cp313-cp313-win_amd64.whl (9.7 MB)
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Installing collected packages: numpy, pandas

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Successfully installed numpy-2.4.4 pandas-3.0.2
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import sys
!"{sys.executable}" -m pip install seaborn
```

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Collecting seaborn
  Downloading seaborn-0.13.2-py3-none-any.whl.metadata (5.4 kB)
Requirement already satisfied: numpy!=1.24.0,>=1.20 in D:\Python_3.13\
Lib\site-packages (from seaborn) (2.4.4)
Requirement already satisfied: pandas>=1.2 in D:\Python_3.13\Lib\site-
packages (from seaborn) (3.0.2)
Collecting matplotlib!=3.6.1,>=3.4 (from seaborn)
  Downloading matplotlib-3.10.8-cp313-cp313-win_amd64.whl.metadata (52
kB)
Collecting contourpy>=1.0.1 (from matplotlib!=3.6.1,>=3.4->seaborn)
  Downloading contourpy-1.3.3-cp313-cp313-win_amd64.whl.metadata (5.5
kB)
Collecting cycler>=0.10 (from matplotlib!=3.6.1,>=3.4->seaborn)
  Downloading cycler-0.12.1-py3-none-any.whl.metadata (3.8 kB)
Collecting fonttools>=4.22.0 (from matplotlib!=3.6.1,>=3.4->seaborn)
  Downloading fonttools-4.62.1-cp313-cp313-win_amd64.whl.metadata (119
kB)
Collecting kiwisolver>=1.3.1 (from matplotlib!=3.6.1,>=3.4->seaborn)
  Downloading kiwisolver-1.5.0-cp313-cp313-win_amd64.whl.metadata (5.2
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Requirement already satisfied: packaging>=20.0 in D:\Python_3.13\Lib\
site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (26.0)
Collecting pillow>=8 (from matplotlib!=3.6.1,>=3.4->seaborn)
  Downloading pillow-12.2.0-cp313-cp313-win_amd64.whl.metadata (9.0
kB)
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Collecting pyparsing>=3 (from matplotlib!=3.6.1,>=3.4->seaborn)
  Downloading pyparsing-3.3.2-py3-none-any.whl.metadata (5.8 kB)
Requirement already satisfied: python-dateutil>=2.7 in D:\Python_3.13\
Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn)
(2.9.0.post0)
Requirement already satisfied: tzdata in D:\Python_3.13\Lib\site-
packages (from pandas>=1.2->seaborn) (2026.1)
Requirement already satisfied: six>=1.5 in D:\Python_3.13\Lib\site-
packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3.4->seaborn)
(1.17.0)
Downloading seaborn-0.13.2-py3-none-any.whl (294 kB)
Downloading matplotlib-3.10.8-cp313-cp313-win_amd64.whl (8.1 MB)
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Downloading contourpy-1.3.3-cp313-cp313-win_amd64.whl (226 kB)
Downloading cycler-0.12.1-py3-none-any.whl (8.3 kB)
Downloading fonttools-4.62.1-cp313-cp313-win_amd64.whl (2.3 MB)
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Downloading pillow-12.2.0-cp313-cp313-win_amd64.whl (7.1 MB)
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Downloading pyparsing-3.3.2-py3-none-any.whl (122 kB)
Installing collected packages: pyparsing, pillow, kiwisolver,
fonttools, cycler, contourpy, matplotlib, seaborn

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Successfully installed contourpy-1.3.3 cycler-0.12.1 fonttools-4.62.1
kiwisolver-1.5.0 matplotlib-3.10.8 pillow-12.2.0 pyparsing-3.3.2
seaborn-0.13.2
```

```
import sys
!"{sys.executable}" -m pip install -U scikit-learn
```

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Collecting scikit-learn
  Downloading scikit_learn-1.8.0-cp313-cp313-win_amd64.whl.metadata
(11 kB)
Requirement already satisfied: numpy>=1.24.1 in D:\Python_3.13\Lib\
site-packages (from scikit-learn) (2.4.4)
Collecting scipy>=1.10.0 (from scikit-learn)
  Downloading scipy-1.17.1-cp313-cp313-win_amd64.whl.metadata (60 kB)
Collecting joblib>=1.3.0 (from scikit-learn)
  Downloading joblib-1.5.3-py3-none-any.whl.metadata (5.5 kB)
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Collecting threadpoolctl>=3.2.0 (from scikit-learn)
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Downloading scikit_learn-1.8.0-cp313-cp313-win_amd64.whl (8.0 MB)
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Downloading scipy-1.17.1-cp313-cp313-win_amd64.whl (36.5 MB)
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Downloading threadpoolctl-3.6.0-py3-none-any.whl (18 kB)
Installing collected packages: threadpoolctl, scipy, joblib, scikit-
learn

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Successfully installed joblib-1.5.3 scikit-learn-1.8.0 scipy-1.17.1
threadpoolctl-3.6.0
```

1. Develop a program to create histograms for all numerical features and analyze the distribution of each feature. Generate box plots for all numerical features and identify any outliers. Use the California Housing dataset.

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.datasets import fetch_california_housing

# Step 1: Load the California Housing dataset
data = fetch_california_housing (as_frame = True)
housing_df = data.frame

# Step 2: Create histograms for numerical features
numerical_features = housing_df.select_dtypes (include =
[np.number]).columns

# Plot and save histograms
plt.figure (figsize = (15, 10))
for i, feature in enumerate (numerical_features) :
    plt.subplot (3, 3, i + 1)
    sns.histplot (housing_df [feature], kde = True, bins = 30, color
='blue')
    plt.title (f'Distribution of {feature}')
plt.tight_layout ()

# SAVE the histogram plot
plt.savefig ('california_histograms.png', dpi = 300, bbox_inches
='tight')
print ("Histograms saved as 'california_histograms.png'")
plt.show ()

# Step 3: Generate box plots for numerical features
plt.figure (figsize = (15, 10))
for i, feature in enumerate (numerical_features) :
    plt.subplot (3, 3, i + 1)
```

```
    sns.boxplot (x = housing_df [feature], color = 'orange')
    plt.title (f'Box Plot of {feature}')
    plt.tight_layout ()

# SAVE the boxplot
plt.savefig ('california_boxplots.png', dpi = 300, bbox_inches
='tight')
print ("Boxplots saved as 'california_boxplots.png'")
plt.show ()

# Step 4: Identify outliers using the IQR method
print ("\nOutliers Detection:")
outliers_summary = {}
for feature in numerical_features :
    Q1 = housing_df [feature].quantile (0.25)
    Q3 = housing_df [feature].quantile (0.75)

    IQR = Q3 - Q1

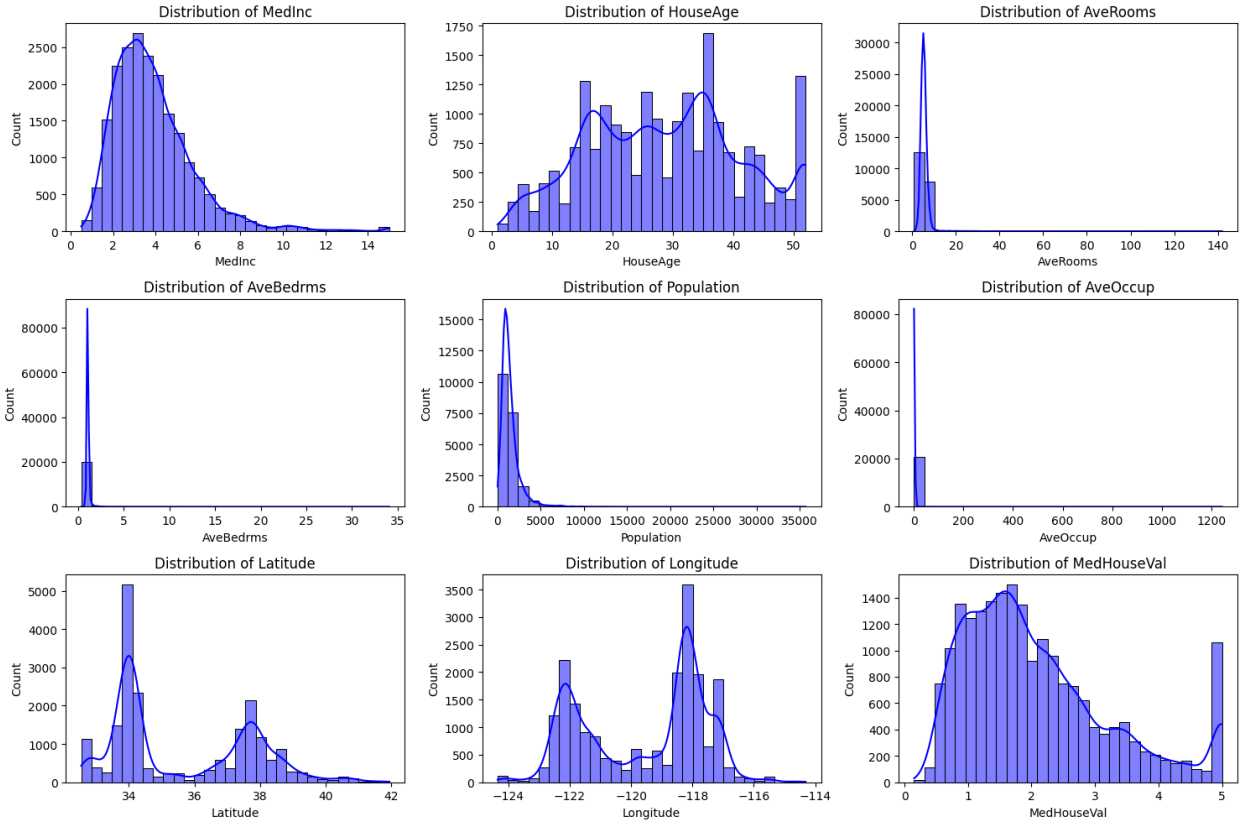
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    outliers = housing_df [(housing_df [feature] < lower_bound) |
(housing_df [feature] > upper_bound)]
    outliers_summary [feature] = len (outliers)

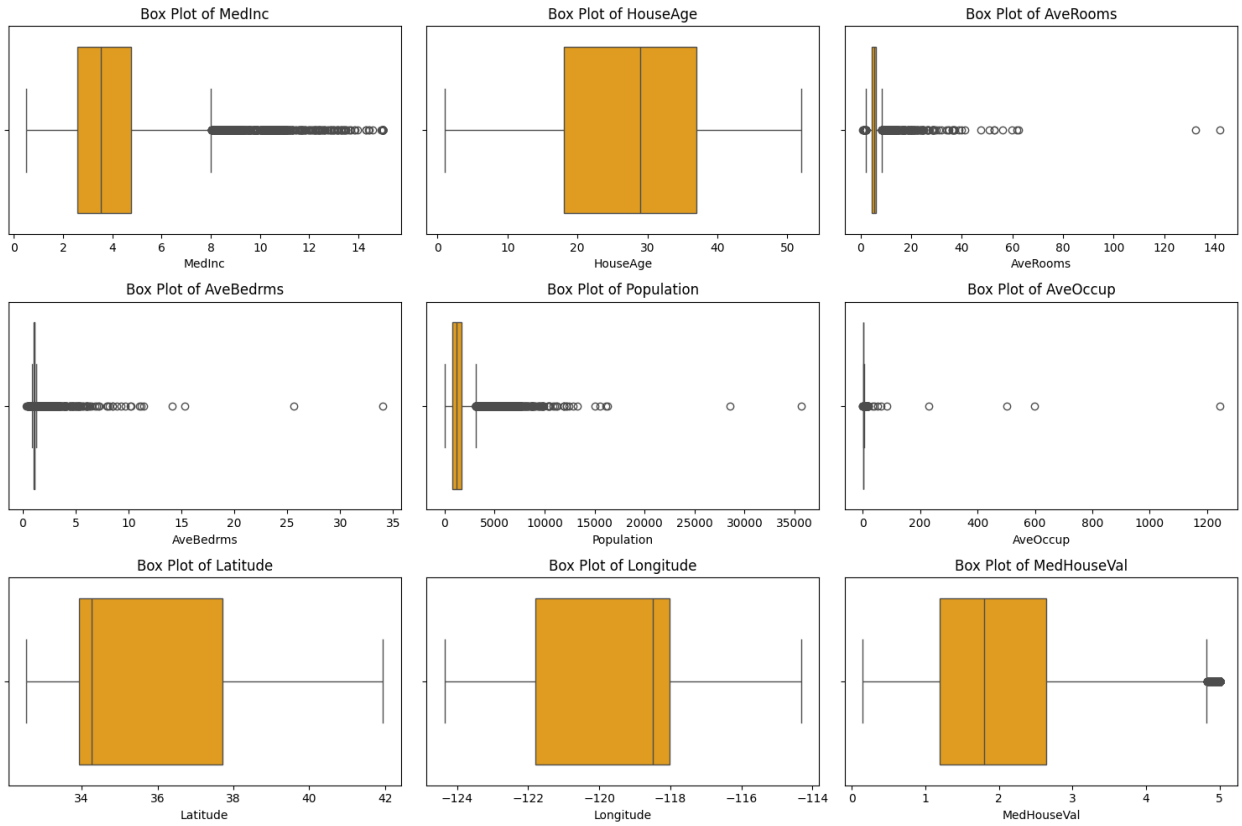
    print (f"{feature} : {len (outliers)} outliers")

# Optional: Print a summary of the dataset
print ("\nDataset Summary:")
print (housing_df.describe ())
```

```
Histograms saved as 'california_histograms.png'
```




```
Boxplots saved as 'california_boxplots.png'
```




```
Outliers Detection:
MedInc : 681 outliers
HouseAge : 0 outliers
AveRooms : 511 outliers
AveBedrms : 1424 outliers
Population : 1196 outliers
AveOccup : 711 outliers
Latitude : 0 outliers
Longitude : 0 outliers
MedHouseVal : 1071 outliers

Dataset Summary:
             MedInc      HouseAge      AveRooms     AveBedrms
Population  \
count  20640.000000  20640.000000  20640.000000  20640.000000
20640.000000
mean       3.870671     28.639486      5.429000      1.096675
1425.476744
std        1.899822     12.585558      2.474173      0.473911
1132.462122
min        0.499900      1.000000      0.846154      0.333333
3.000000
25%        2.563400     18.000000      4.440716      1.006079
```

```
787.000000
50%        3.534800     29.000000      5.229129      1.048780
1166.000000
75%        4.743250     37.000000      6.052381      1.099526
1725.000000
max       15.000100     52.000000    141.909091     34.066667
35682.000000

           AveOccup      Latitude     Longitude   MedHouseVal
count  20640.000000  20640.000000  20640.000000  20640.000000
mean       3.070655     35.631861   -119.569704      2.068558
std       10.386050      2.135952      2.003532      1.153956
min        0.692308     32.540000   -124.350000      0.149990
25%        2.429741     33.930000   -121.800000      1.196000
50%        2.818116     34.260000   -118.490000      1.797000
75%        3.282261     37.710000   -118.010000      2.647250
max     1243.333333     41.950000   -114.310000      5.000010
```

2. Develop a program to compute the correlation matrix to understand the relationships between pairs of features. Visualize the correlation matrix using a heatmap to know which variables have strong positive/negative correlations. Create a pair plot to visualize pairwise relationships between features. Use California Housing dataset.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import fetch_california_housing

# Step 1: Load the California Housing dataset
california_data = fetch_california_housing (as_frame = True)
data = california_data.frame

#Step 2: Try data correlation matrix
correlation_matrix = data.corr ()

# Step 3: Plot and save the heatmap
```

```
plt.figure (figsize = (10, 8))
sns.heatmap (correlation_matrix, annot = True, cmap = 'coolwarm',
linewidth = 0.5)

# Save the heatmap
plt.savefig('california_heatmap.png', dpi=300, bbox_inches='tight')
print("Heatmaps saved as 'california_heatmap.png'")

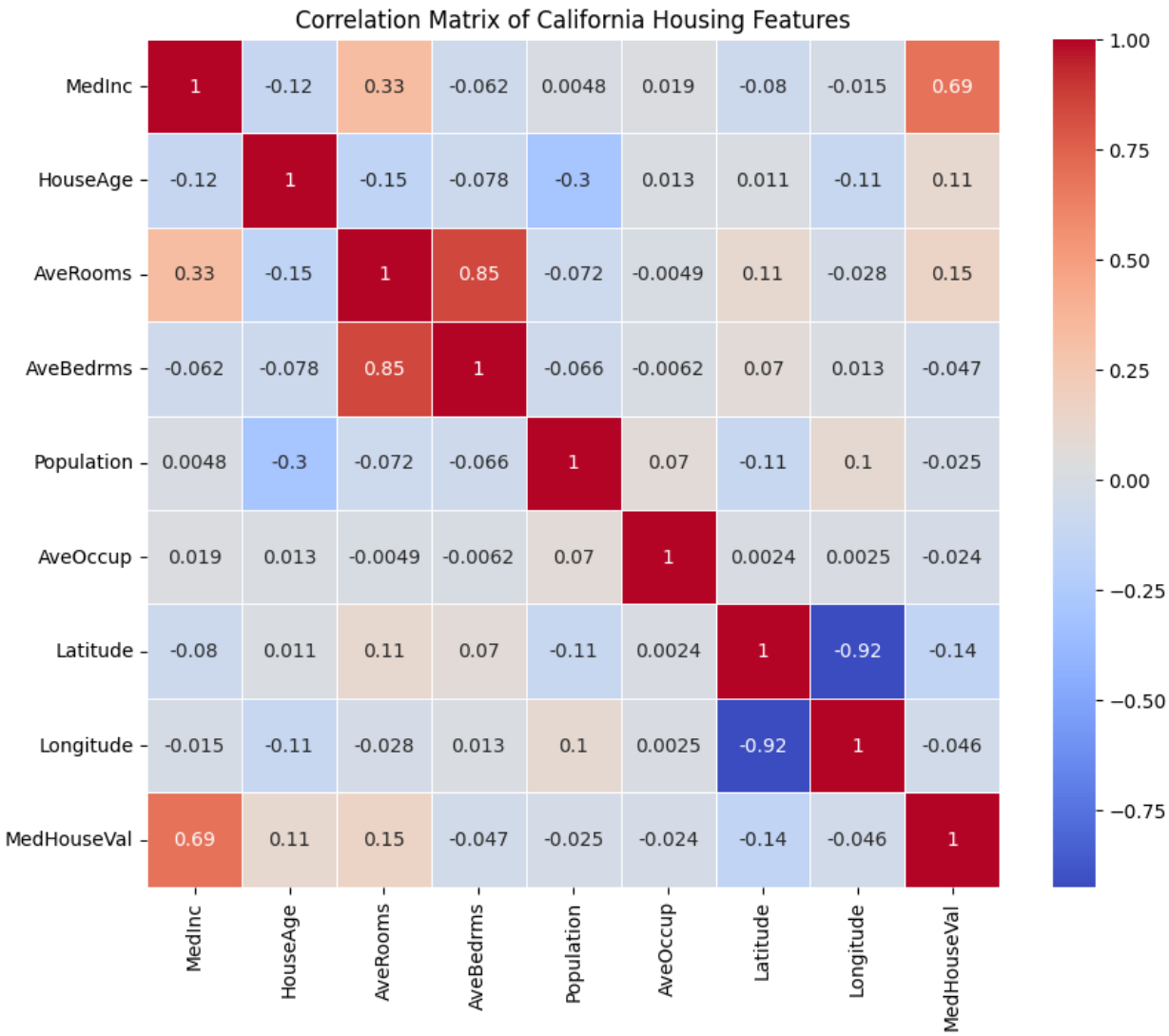
# Show the heatmap
plt.title ('Correlation Matrix of California Housing Features')
plt.show ()

# Step 4: Plot and save the pairplot
sns.pairplot (data, diag_kind = 'kde', plot_kws = {'alpha' : 0.5})

# Save the pairplot
plt.savefig('california_pairplot.png', dpi=300, bbox_inches='tight')
print("Pairplots saved as 'california_pairplot.png'")

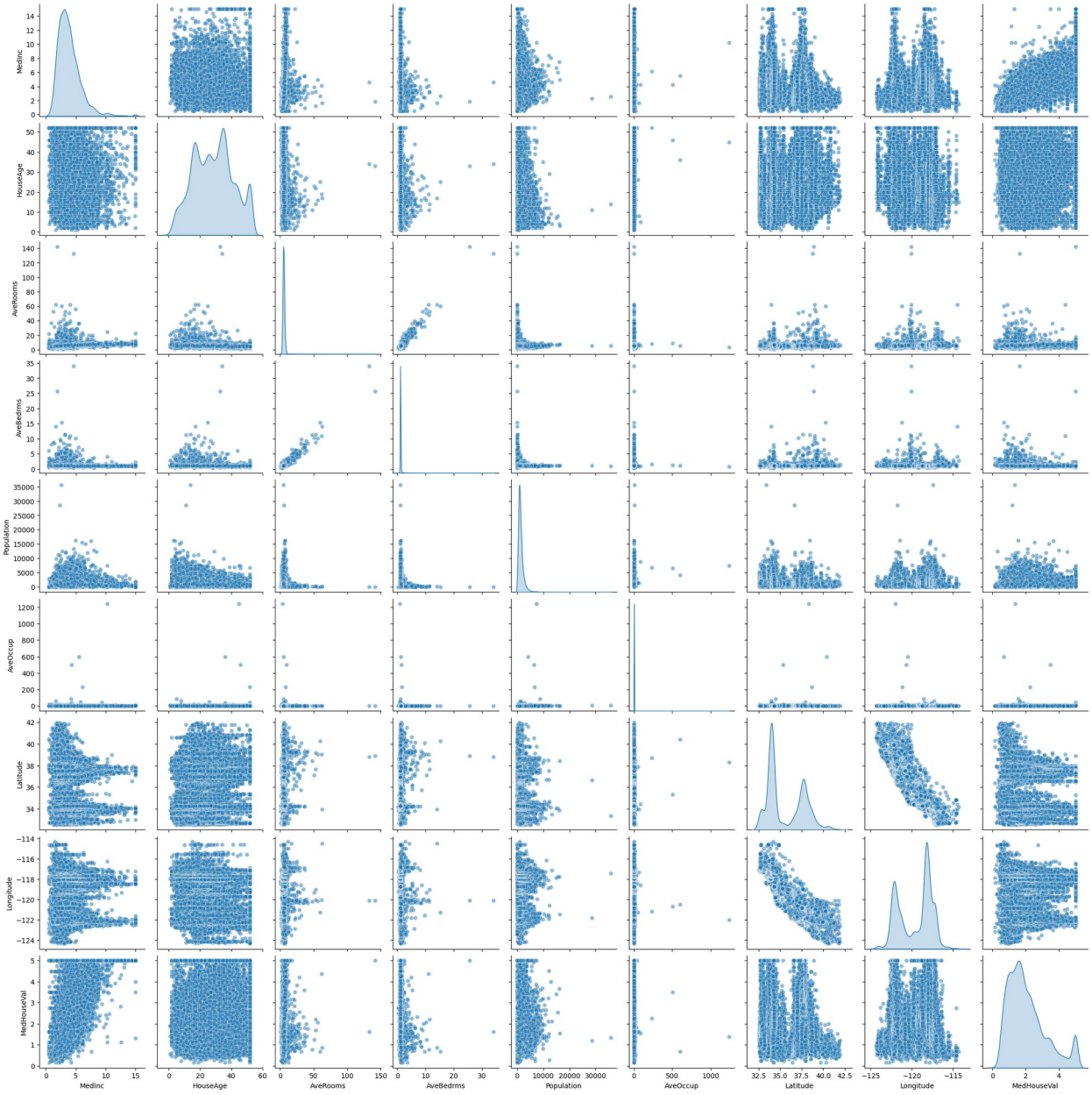
# Show the pairplot
plt.suptitle ('Pair Plot of California  Housing Features', y = 1.02)
plt.show ()
```

```
Heatmaps saved as 'california_heatmap.png'
```




```
Pairplots saved as 'california_pairplot.png'
```

Pair Plot of California Housing Features



3. Develop a program to implement Principal Component Analysis (PCA) for reducing the dimensionality of the Iris dataset from 4 features to 2.

```
import numpy as np
import pandas as pd
from sklearn.datasets import load_iris
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt

# Load the Iris dataset
iris = load_iris ()
data = iris.data
labels = iris.target
label_names = iris.target_names

# Convert to a DataFrame for better vizualization
iris_df = pd.DataFrame (data, columns = iris.feature_names)

# Perform PCA to reduce dimensionality to 2
pca = PCA (n_components = 2)
data_reduced = pca.fit_transform (data)

# Create a DataFrame for the reduced data
reduced_df = pd.DataFrame (data_reduced, columns = ['Principal
Component 1', 'Principal Component 2'])
reduced_df ['Label'] = labels

# Plot the reduced data
plt.figure (figsize = (8, 6))
colors = ['r', 'g', 'b']

for i, label in enumerate (np.unique (labels)) :
    plt.scatter (
        reduced_df [reduced_df ['Label'] == label] ['Principal
Component 1'],
        reduced_df [reduced_df ['Label'] == label] ['Principal
Component 2'],
        label = label_names [label],
        color = colors [i]
    )

# Display the context of the PCA
plt.title ('PCA on Iris Dataset')

# Display the axes
```

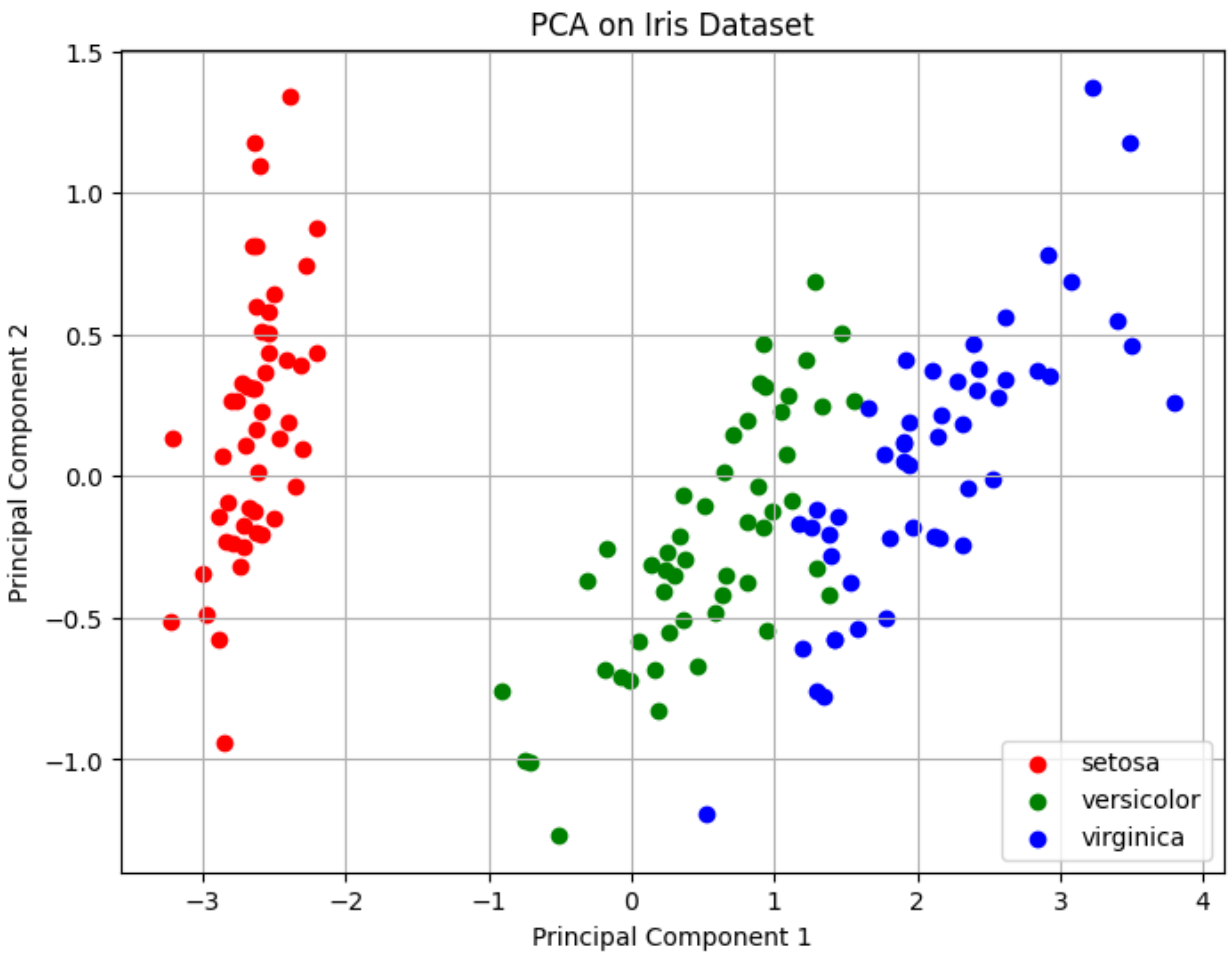
```
plt.xlabel ('Principal Component 1')
plt.ylabel ('Principal Component 2')

# Show the view
plt.legend ()
plt.grid ()

plt.savefig ('pca_iris_dataset.png', dpi = 300, bbox_inches = 'tight')

plt.show ()

# Save the PCA
print ("PCA saved as 'pca_iris_dataset.png'")
```




```
PCA saved as 'pca_iris_dataset.png'
```

4. For a given set of training data examples stored in a .csv file, implement and demonstrate the Find-S algorithm to output a description of the set of all hypotheses consistent with the training examples.

```
import pandas as pd

def find_s_algorithm (file_path) :
    data = pd.read_csv (file_path) # include 'training_data.csv'
inside the same folder as in your workspace

    print ("Training data: ", data)

    attributes = data.columns [:-1]
    class_label = data.columns [-1]

    hypothesis = ['?' for _ in attributes]

    for index, row in data.iterrows () :
        if row [class_label] == 'Yes' :
            for i, value in enumerate (row [attributes]) :
                if hypothesis [i] == '?' or hypothesis [i] == value :
hypothesis [i] = value

                else : hypothesis[i] = '?'
    return hypothesis

file_path = 'training_data.csv'
hypothesis = find_s_algorithm (file_path)

print ("\nThe final hypothesis is: ", hypothesis)
```

```
Training data:      Outlook Temperature Humidity  Windy PlayTennis
0     Sunny         Hot     High  False         No
1     Sunny         Hot     High   True         No
2  Overcast         Hot     High  False        Yes
3      Rain        Cold     High  False        Yes
4      Rain        Cold     High   True         No
5  Overcast         Hot     High   True        Yes
6     Sunny         Hot     High  False         No

The final hypothesis is:  ['Overcast', 'Hot', 'High', '?']
```

5. Develop a program to implement k-Nearest Neighbour algorithm to classify the randomly generated 100 values of x in the range of [0,1].

Perform the following based on dataset generated.

a) Label the first 50 points {x1, ……, x50} as follows:

if (xi ≤ 0.5), then xi Class1, else xi ∈ Class1

b) Classify the remaining points, x51, ……, x100 using KNN. Perform this for k = 1, 2, 3, 4, 5, 20, 30.

```
import numpy as np
import matplotlib.pyplot as plt
from collections import Counter

data = np.random.rand (100)
labels = ["Class1" if x <= 0.5 else "Class2" for x in data [:50]]

def euclidean_distance (x1, x2) : return abs (x1 - x2)

def knn_classifier (train_data, train_labels, test_point, k):
    distances = [(euclidean_distance (test_point, train_data[i]),
    train_labels[i]) for i in range (len (train_data) )]

    distances.sort (key = lambda x : x [0])

    k_nearest_neighbors = distances [:k]
    k_nearest_labels = [label for _, label in k_nearest_neighbors]

    return Counter (k_nearest_labels).most_common (1) [0] [0]
```

```
train_data = data [:50]
train_labels = labels
test_data = data [50:]
k_values = [1, 2, 3, 4, 5, 20, 30]

print ("--- k-nearest Neighbors Classification ---")
print ("Training dataset: First 50 points labelled based on the rule
(x <= 0.5 -> Class1, x > 0.5 -> Class2)")
print ("Testing dataset: Remaining 50 points to be classified\n")

results = {}

for k in k_values :
    print (f"Results for k = {k}:")

    classified_labels = [knn_classifier (train_data, train_labels,
test_point, k)
                         for test_point in test_data]
    results [k] = classified_labels

    for i, label in enumerate (classified_labels, start = 51) :
        print (f"Point x {i} (value : {test_data [i - 51] : .4f}) is
classified as {label}")

    print ("\n")

print ("Classification complete.\n")

for k in k_values :
    classified_labels = results [k]

    class1_points = [test_data [i] for i in range (len (test_data) )

if classified_labels [i] == "Class1"]
    class2_points = [test_data [i] for i in range (len (test_data) )
if classified_labels [i] == "Class2"]
    plt.figure (figsize = (10, 6))

    plt.scatter (
        train_data,
        [0] * len (train_data),
        c = ["blue" if label == "Class1" else "red" for label in
train_labels],
        label = "Training Data",
        marker = "o"
    )

    plt.scatter (
        class1_points,
```

```
        [1] * len (class1_points),
        c = "blue",
        label = "Class1 (Test)",
        marker = "x"
    )

    plt.scatter (
        class2_points,
        [1] * len (class2_points),
        c = "red",
        label = "Class2 (Test)",
        marker = "x"
    )

    plt.title (f"k-NN Classification Results for k = {k}")

    plt.xlabel ("Data Points")
    plt.ylabel ("Classification Level")

    plt.legend ()
    plt.grid ()

    plt.savefig ('knn.png', dpi = 300, bbox_inches = 'tight')

    plt.show ()
```

```
--- k-nearest Neighbors Classification ---
Training dataset: First 50 points labelled based on the rule (x <= 0.5
-> Class1, x > 0.5 -> Class2)
Testing dataset: Remaining 50 points to be classified

Results for k = 1:
Point x 51 (value :  0.0330) is classified as Class1
Point x 52 (value :  0.5104) is classified as Class2
Point x 53 (value :  0.2581) is classified as Class1
Point x 54 (value :  0.2446) is classified as Class1
Point x 55 (value :  0.8474) is classified as Class2
Point x 56 (value :  0.5729) is classified as Class2
Point x 57 (value :  0.7938) is classified as Class2
Point x 58 (value :  0.0887) is classified as Class1
Point x 59 (value :  0.1377) is classified as Class1
Point x 60 (value :  0.4370) is classified as Class1
Point x 61 (value :  0.4371) is classified as Class1
Point x 62 (value :  0.8933) is classified as Class2
Point x 63 (value :  0.6128) is classified as Class2
Point x 64 (value :  0.4816) is classified as Class2
Point x 65 (value :  0.8998) is classified as Class2
Point x 66 (value :  0.4412) is classified as Class1
Point x 67 (value :  0.4545) is classified as Class1
Point x 68 (value :  0.3484) is classified as Class1
```

```
Point x 69 (value :  0.2543) is classified as Class1
Point x 70 (value :  0.9073) is classified as Class2
Point x 71 (value :  0.5548) is classified as Class2
Point x 72 (value :  0.8918) is classified as Class2
Point x 73 (value :  0.2346) is classified as Class1
Point x 74 (value :  0.9864) is classified as Class2
Point x 75 (value :  0.1455) is classified as Class1
Point x 76 (value :  0.2170) is classified as Class1
Point x 77 (value :  0.5356) is classified as Class2
Point x 78 (value :  0.8669) is classified as Class2
Point x 79 (value :  0.7231) is classified as Class2
Point x 80 (value :  0.3543) is classified as Class1
Point x 81 (value :  0.7346) is classified as Class2
Point x 82 (value :  0.4147) is classified as Class1
Point x 83 (value :  0.3320) is classified as Class1
Point x 84 (value :  0.0376) is classified as Class1
Point x 85 (value :  0.6599) is classified as Class2
Point x 86 (value :  0.8521) is classified as Class2
Point x 87 (value :  0.7628) is classified as Class2
Point x 88 (value :  0.7053) is classified as Class2
Point x 89 (value :  0.2535) is classified as Class1
Point x 90 (value :  0.5581) is classified as Class2
Point x 91 (value :  0.5090) is classified as Class2
Point x 92 (value :  0.1859) is classified as Class1
Point x 93 (value :  0.1852) is classified as Class1
Point x 94 (value :  0.0245) is classified as Class1
Point x 95 (value :  0.1084) is classified as Class1
Point x 96 (value :  0.6561) is classified as Class2
Point x 97 (value :  0.9366) is classified as Class2
Point x 98 (value :  0.1635) is classified as Class1
Point x 99 (value :  0.8427) is classified as Class2
Point x 100 (value :  0.4878) is classified as Class2


Results for k = 2:
Point x 51 (value :  0.0330) is classified as Class1
Point x 52 (value :  0.5104) is classified as Class2
Point x 53 (value :  0.2581) is classified as Class1
Point x 54 (value :  0.2446) is classified as Class1
Point x 55 (value :  0.8474) is classified as Class2
Point x 56 (value :  0.5729) is classified as Class2
Point x 57 (value :  0.7938) is classified as Class2
Point x 58 (value :  0.0887) is classified as Class1
Point x 59 (value :  0.1377) is classified as Class1
Point x 60 (value :  0.4370) is classified as Class1
Point x 61 (value :  0.4371) is classified as Class1
Point x 62 (value :  0.8933) is classified as Class2
Point x 63 (value :  0.6128) is classified as Class2
Point x 64 (value :  0.4816) is classified as Class2
```

```
Point x 65 (value :  0.8998) is classified as Class2
Point x 66 (value :  0.4412) is classified as Class1
Point x 67 (value :  0.4545) is classified as Class1
Point x 68 (value :  0.3484) is classified as Class1
Point x 69 (value :  0.2543) is classified as Class1
Point x 70 (value :  0.9073) is classified as Class2
Point x 71 (value :  0.5548) is classified as Class2
Point x 72 (value :  0.8918) is classified as Class2
Point x 73 (value :  0.2346) is classified as Class1
Point x 74 (value :  0.9864) is classified as Class2
Point x 75 (value :  0.1455) is classified as Class1
Point x 76 (value :  0.2170) is classified as Class1
Point x 77 (value :  0.5356) is classified as Class2
Point x 78 (value :  0.8669) is classified as Class2
Point x 79 (value :  0.7231) is classified as Class2
Point x 80 (value :  0.3543) is classified as Class1
Point x 81 (value :  0.7346) is classified as Class2
Point x 82 (value :  0.4147) is classified as Class1
Point x 83 (value :  0.3320) is classified as Class1
Point x 84 (value :  0.0376) is classified as Class1
Point x 85 (value :  0.6599) is classified as Class2
Point x 86 (value :  0.8521) is classified as Class2
Point x 87 (value :  0.7628) is classified as Class2
Point x 88 (value :  0.7053) is classified as Class2
Point x 89 (value :  0.2535) is classified as Class1
Point x 90 (value :  0.5581) is classified as Class2
Point x 91 (value :  0.5090) is classified as Class2
Point x 92 (value :  0.1859) is classified as Class1
Point x 93 (value :  0.1852) is classified as Class1
Point x 94 (value :  0.0245) is classified as Class1
Point x 95 (value :  0.1084) is classified as Class1
Point x 96 (value :  0.6561) is classified as Class2
Point x 97 (value :  0.9366) is classified as Class2
Point x 98 (value :  0.1635) is classified as Class1
Point x 99 (value :  0.8427) is classified as Class2
Point x 100 (value :  0.4878) is classified as Class2


Results for k = 3:
Point x 51 (value :  0.0330) is classified as Class1
Point x 52 (value :  0.5104) is classified as Class2
Point x 53 (value :  0.2581) is classified as Class1
Point x 54 (value :  0.2446) is classified as Class1
Point x 55 (value :  0.8474) is classified as Class2
Point x 56 (value :  0.5729) is classified as Class2
Point x 57 (value :  0.7938) is classified as Class2
Point x 58 (value :  0.0887) is classified as Class1
Point x 59 (value :  0.1377) is classified as Class1
Point x 60 (value :  0.4370) is classified as Class1
```

```
Point x 61 (value :  0.4371) is classified as Class1
Point x 62 (value :  0.8933) is classified as Class2
Point x 63 (value :  0.6128) is classified as Class2
Point x 64 (value :  0.4816) is classified as Class2
Point x 65 (value :  0.8998) is classified as Class2
Point x 66 (value :  0.4412) is classified as Class1
Point x 67 (value :  0.4545) is classified as Class1
Point x 68 (value :  0.3484) is classified as Class1
Point x 69 (value :  0.2543) is classified as Class1
Point x 70 (value :  0.9073) is classified as Class2
Point x 71 (value :  0.5548) is classified as Class2
Point x 72 (value :  0.8918) is classified as Class2
Point x 73 (value :  0.2346) is classified as Class1
Point x 74 (value :  0.9864) is classified as Class2
Point x 75 (value :  0.1455) is classified as Class1
Point x 76 (value :  0.2170) is classified as Class1
Point x 77 (value :  0.5356) is classified as Class2
Point x 78 (value :  0.8669) is classified as Class2
Point x 79 (value :  0.7231) is classified as Class2
Point x 80 (value :  0.3543) is classified as Class1
Point x 81 (value :  0.7346) is classified as Class2
Point x 82 (value :  0.4147) is classified as Class1
Point x 83 (value :  0.3320) is classified as Class1
Point x 84 (value :  0.0376) is classified as Class1
Point x 85 (value :  0.6599) is classified as Class2
Point x 86 (value :  0.8521) is classified as Class2
Point x 87 (value :  0.7628) is classified as Class2
Point x 88 (value :  0.7053) is classified as Class2
Point x 89 (value :  0.2535) is classified as Class1
Point x 90 (value :  0.5581) is classified as Class2
Point x 91 (value :  0.5090) is classified as Class2
Point x 92 (value :  0.1859) is classified as Class1
Point x 93 (value :  0.1852) is classified as Class1
Point x 94 (value :  0.0245) is classified as Class1
Point x 95 (value :  0.1084) is classified as Class1
Point x 96 (value :  0.6561) is classified as Class2
Point x 97 (value :  0.9366) is classified as Class2
Point x 98 (value :  0.1635) is classified as Class1
Point x 99 (value :  0.8427) is classified as Class2
Point x 100 (value :  0.4878) is classified as Class2


Results for k = 4:
Point x 51 (value :  0.0330) is classified as Class1
Point x 52 (value :  0.5104) is classified as Class2
Point x 53 (value :  0.2581) is classified as Class1
Point x 54 (value :  0.2446) is classified as Class1
Point x 55 (value :  0.8474) is classified as Class2
Point x 56 (value :  0.5729) is classified as Class2
```

```
Point x 57 (value :  0.7938) is classified as Class2
Point x 58 (value :  0.0887) is classified as Class1
Point x 59 (value :  0.1377) is classified as Class1
Point x 60 (value :  0.4370) is classified as Class1
Point x 61 (value :  0.4371) is classified as Class1
Point x 62 (value :  0.8933) is classified as Class2
Point x 63 (value :  0.6128) is classified as Class2
Point x 64 (value :  0.4816) is classified as Class2
Point x 65 (value :  0.8998) is classified as Class2
Point x 66 (value :  0.4412) is classified as Class1
Point x 67 (value :  0.4545) is classified as Class1
Point x 68 (value :  0.3484) is classified as Class1
Point x 69 (value :  0.2543) is classified as Class1
Point x 70 (value :  0.9073) is classified as Class2
Point x 71 (value :  0.5548) is classified as Class2
Point x 72 (value :  0.8918) is classified as Class2
Point x 73 (value :  0.2346) is classified as Class1
Point x 74 (value :  0.9864) is classified as Class2
Point x 75 (value :  0.1455) is classified as Class1
Point x 76 (value :  0.2170) is classified as Class1
Point x 77 (value :  0.5356) is classified as Class2
Point x 78 (value :  0.8669) is classified as Class2
Point x 79 (value :  0.7231) is classified as Class2
Point x 80 (value :  0.3543) is classified as Class1
Point x 81 (value :  0.7346) is classified as Class2
Point x 82 (value :  0.4147) is classified as Class1
Point x 83 (value :  0.3320) is classified as Class1
Point x 84 (value :  0.0376) is classified as Class1
Point x 85 (value :  0.6599) is classified as Class2
Point x 86 (value :  0.8521) is classified as Class2
Point x 87 (value :  0.7628) is classified as Class2
Point x 88 (value :  0.7053) is classified as Class2
Point x 89 (value :  0.2535) is classified as Class1
Point x 90 (value :  0.5581) is classified as Class2
Point x 91 (value :  0.5090) is classified as Class2
Point x 92 (value :  0.1859) is classified as Class1
Point x 93 (value :  0.1852) is classified as Class1
Point x 94 (value :  0.0245) is classified as Class1
Point x 95 (value :  0.1084) is classified as Class1
Point x 96 (value :  0.6561) is classified as Class2
Point x 97 (value :  0.9366) is classified as Class2
Point x 98 (value :  0.1635) is classified as Class1
Point x 99 (value :  0.8427) is classified as Class2
Point x 100 (value :  0.4878) is classified as Class2


Results for k = 5:
Point x 51 (value :  0.0330) is classified as Class1
Point x 52 (value :  0.5104) is classified as Class2
```

```
Point x 53 (value :  0.2581) is classified as Class1
Point x 54 (value :  0.2446) is classified as Class1
Point x 55 (value :  0.8474) is classified as Class2
Point x 56 (value :  0.5729) is classified as Class2
Point x 57 (value :  0.7938) is classified as Class2
Point x 58 (value :  0.0887) is classified as Class1
Point x 59 (value :  0.1377) is classified as Class1
Point x 60 (value :  0.4370) is classified as Class1
Point x 61 (value :  0.4371) is classified as Class1
Point x 62 (value :  0.8933) is classified as Class2
Point x 63 (value :  0.6128) is classified as Class2
Point x 64 (value :  0.4816) is classified as Class2
Point x 65 (value :  0.8998) is classified as Class2
Point x 66 (value :  0.4412) is classified as Class1
Point x 67 (value :  0.4545) is classified as Class1
Point x 68 (value :  0.3484) is classified as Class1
Point x 69 (value :  0.2543) is classified as Class1
Point x 70 (value :  0.9073) is classified as Class2
Point x 71 (value :  0.5548) is classified as Class2
Point x 72 (value :  0.8918) is classified as Class2
Point x 73 (value :  0.2346) is classified as Class1
Point x 74 (value :  0.9864) is classified as Class2
Point x 75 (value :  0.1455) is classified as Class1
Point x 76 (value :  0.2170) is classified as Class1
Point x 77 (value :  0.5356) is classified as Class2
Point x 78 (value :  0.8669) is classified as Class2
Point x 79 (value :  0.7231) is classified as Class2
Point x 80 (value :  0.3543) is classified as Class1
Point x 81 (value :  0.7346) is classified as Class2
Point x 82 (value :  0.4147) is classified as Class1
Point x 83 (value :  0.3320) is classified as Class1
Point x 84 (value :  0.0376) is classified as Class1
Point x 85 (value :  0.6599) is classified as Class2
Point x 86 (value :  0.8521) is classified as Class2
Point x 87 (value :  0.7628) is classified as Class2
Point x 88 (value :  0.7053) is classified as Class2
Point x 89 (value :  0.2535) is classified as Class1
Point x 90 (value :  0.5581) is classified as Class2
Point x 91 (value :  0.5090) is classified as Class2
Point x 92 (value :  0.1859) is classified as Class1
Point x 93 (value :  0.1852) is classified as Class1
Point x 94 (value :  0.0245) is classified as Class1
Point x 95 (value :  0.1084) is classified as Class1
Point x 96 (value :  0.6561) is classified as Class2
Point x 97 (value :  0.9366) is classified as Class2
Point x 98 (value :  0.1635) is classified as Class1
Point x 99 (value :  0.8427) is classified as Class2
Point x 100 (value :  0.4878) is classified as Class2
```

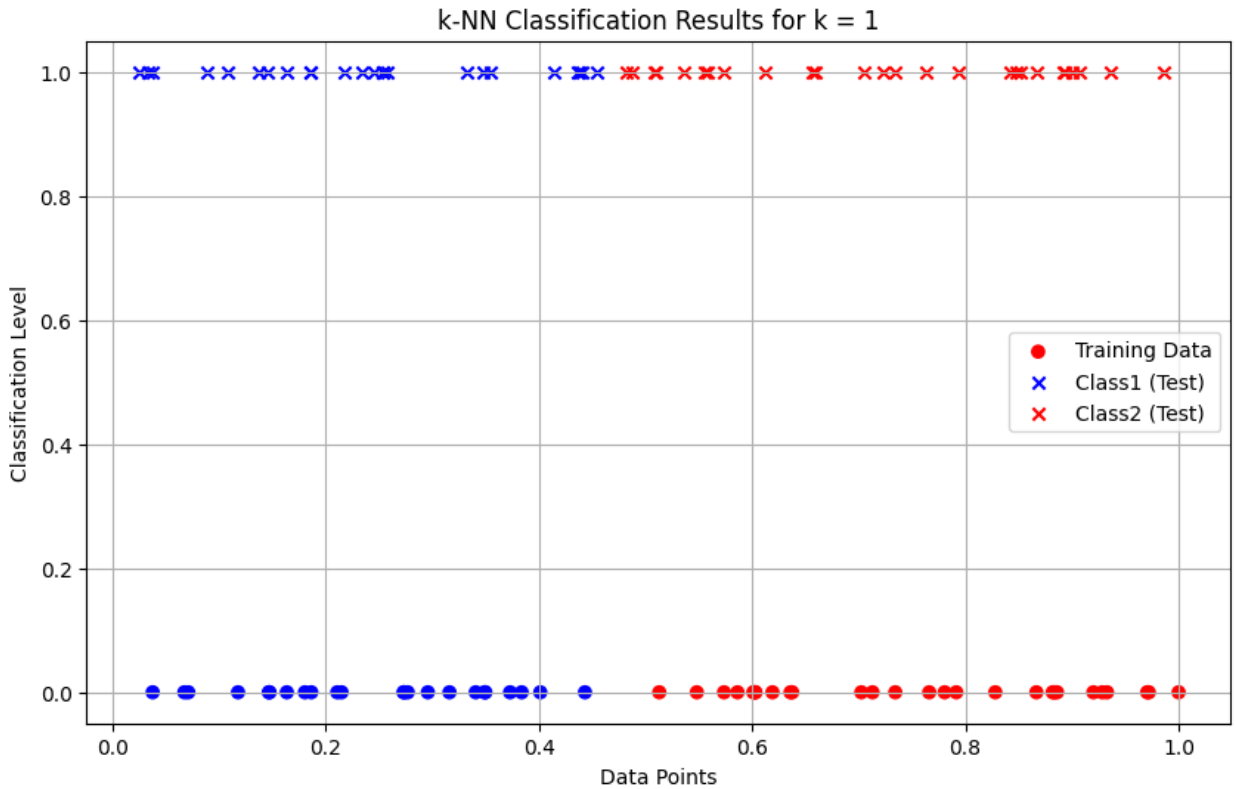
```
Results for k = 20:
Point x 51 (value :  0.0330) is classified as Class1
Point x 52 (value :  0.5104) is classified as Class2
Point x 53 (value :  0.2581) is classified as Class1
Point x 54 (value :  0.2446) is classified as Class1
Point x 55 (value :  0.8474) is classified as Class2
Point x 56 (value :  0.5729) is classified as Class2
Point x 57 (value :  0.7938) is classified as Class2
Point x 58 (value :  0.0887) is classified as Class1
Point x 59 (value :  0.1377) is classified as Class1
Point x 60 (value :  0.4370) is classified as Class1
Point x 61 (value :  0.4371) is classified as Class1
Point x 62 (value :  0.8933) is classified as Class2
Point x 63 (value :  0.6128) is classified as Class2
Point x 64 (value :  0.4816) is classified as Class2
Point x 65 (value :  0.8998) is classified as Class2
Point x 66 (value :  0.4412) is classified as Class1
Point x 67 (value :  0.4545) is classified as Class1
Point x 68 (value :  0.3484) is classified as Class1
Point x 69 (value :  0.2543) is classified as Class1
Point x 70 (value :  0.9073) is classified as Class2
Point x 71 (value :  0.5548) is classified as Class2
Point x 72 (value :  0.8918) is classified as Class2
Point x 73 (value :  0.2346) is classified as Class1
Point x 74 (value :  0.9864) is classified as Class2
Point x 75 (value :  0.1455) is classified as Class1
Point x 76 (value :  0.2170) is classified as Class1
Point x 77 (value :  0.5356) is classified as Class2
Point x 78 (value :  0.8669) is classified as Class2
Point x 79 (value :  0.7231) is classified as Class2
Point x 80 (value :  0.3543) is classified as Class1
Point x 81 (value :  0.7346) is classified as Class2
Point x 82 (value :  0.4147) is classified as Class1
Point x 83 (value :  0.3320) is classified as Class1
Point x 84 (value :  0.0376) is classified as Class1
Point x 85 (value :  0.6599) is classified as Class2
Point x 86 (value :  0.8521) is classified as Class2
Point x 87 (value :  0.7628) is classified as Class2
Point x 88 (value :  0.7053) is classified as Class2
Point x 89 (value :  0.2535) is classified as Class1
Point x 90 (value :  0.5581) is classified as Class2
Point x 91 (value :  0.5090) is classified as Class2
Point x 92 (value :  0.1859) is classified as Class1
Point x 93 (value :  0.1852) is classified as Class1
Point x 94 (value :  0.0245) is classified as Class1
Point x 95 (value :  0.1084) is classified as Class1
Point x 96 (value :  0.6561) is classified as Class2
Point x 97 (value :  0.9366) is classified as Class2
```

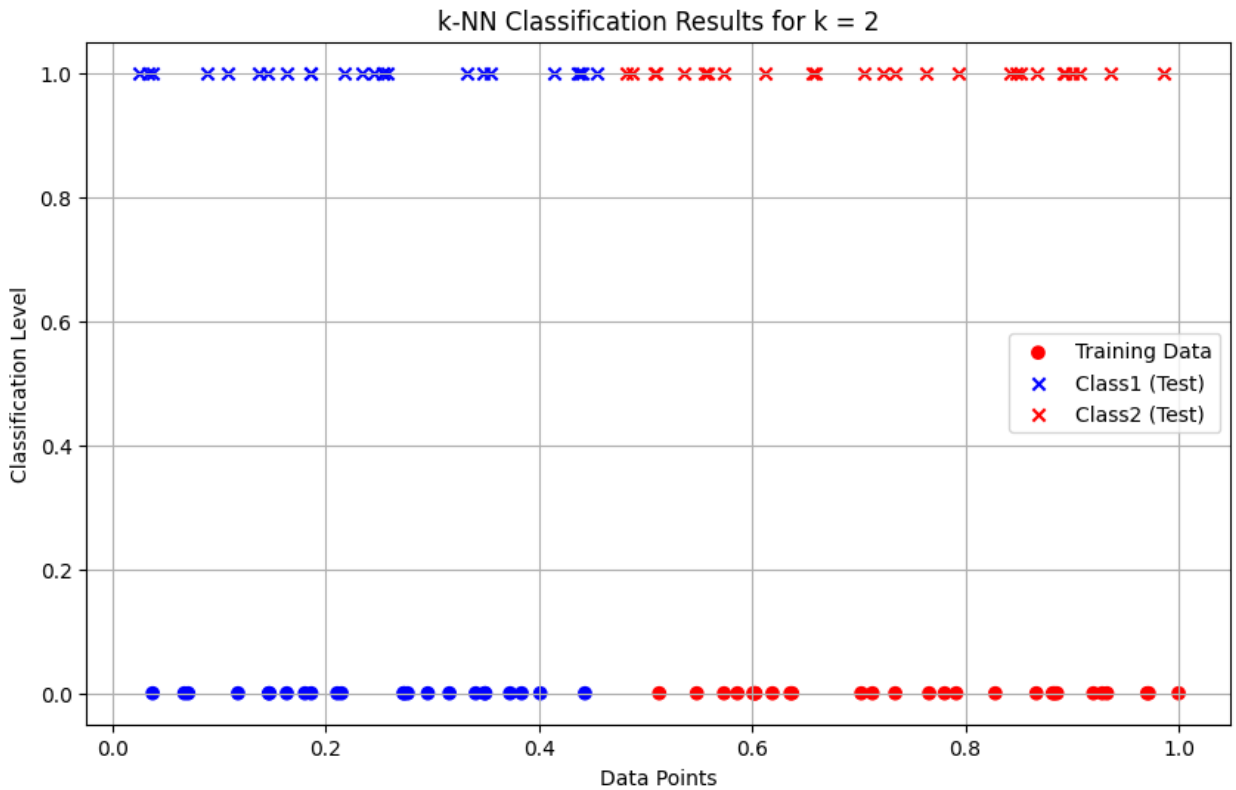
```
Point x 98 (value :  0.1635) is classified as Class1
Point x 99 (value :  0.8427) is classified as Class2
Point x 100 (value :  0.4878) is classified as Class2

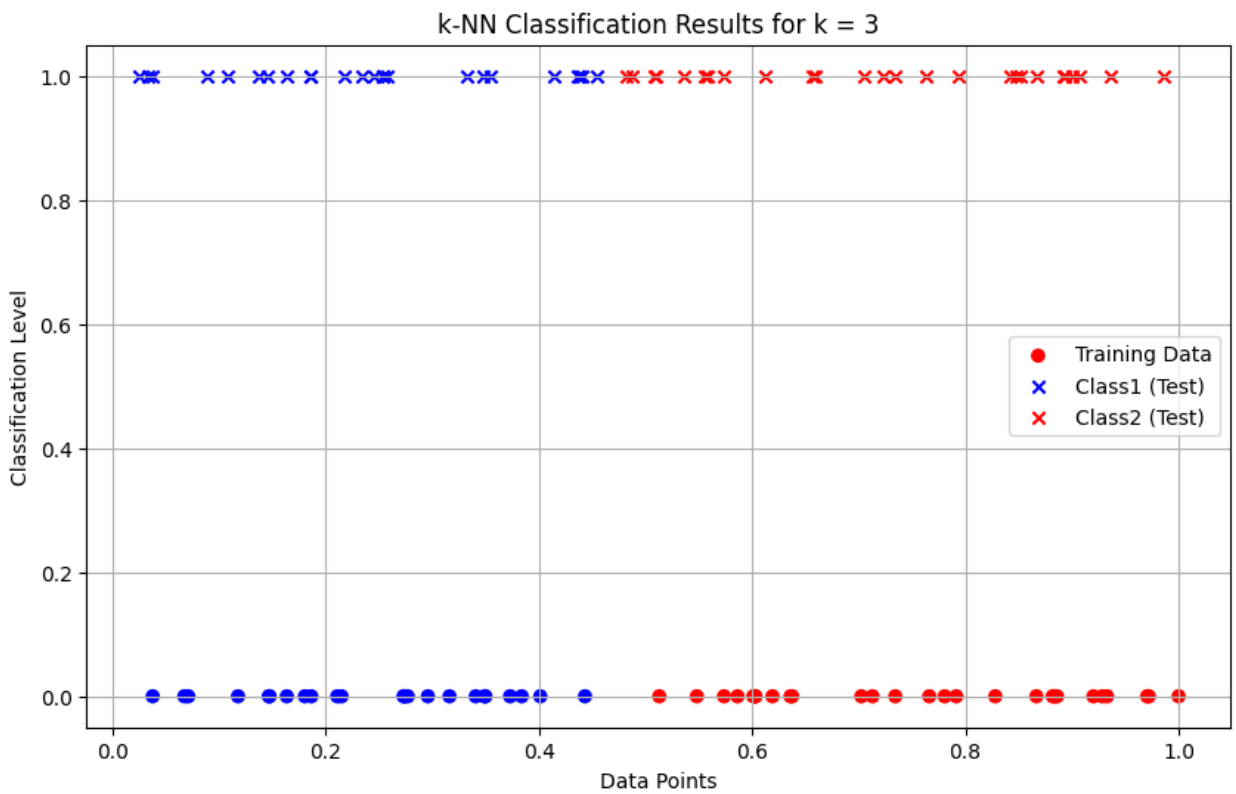

Results for k = 30:
Point x 51 (value :  0.0330) is classified as Class1
Point x 52 (value :  0.5104) is classified as Class2
Point x 53 (value :  0.2581) is classified as Class1
Point x 54 (value :  0.2446) is classified as Class1
Point x 55 (value :  0.8474) is classified as Class2
Point x 56 (value :  0.5729) is classified as Class2
Point x 57 (value :  0.7938) is classified as Class2
Point x 58 (value :  0.0887) is classified as Class1
Point x 59 (value :  0.1377) is classified as Class1
Point x 60 (value :  0.4370) is classified as Class1
Point x 61 (value :  0.4371) is classified as Class1
Point x 62 (value :  0.8933) is classified as Class2
Point x 63 (value :  0.6128) is classified as Class2
Point x 64 (value :  0.4816) is classified as Class2
Point x 65 (value :  0.8998) is classified as Class2
Point x 66 (value :  0.4412) is classified as Class1
Point x 67 (value :  0.4545) is classified as Class1
Point x 68 (value :  0.3484) is classified as Class1
Point x 69 (value :  0.2543) is classified as Class1
Point x 70 (value :  0.9073) is classified as Class2
Point x 71 (value :  0.5548) is classified as Class2
Point x 72 (value :  0.8918) is classified as Class2
Point x 73 (value :  0.2346) is classified as Class1
Point x 74 (value :  0.9864) is classified as Class2
Point x 75 (value :  0.1455) is classified as Class1
Point x 76 (value :  0.2170) is classified as Class1
Point x 77 (value :  0.5356) is classified as Class2
Point x 78 (value :  0.8669) is classified as Class2
Point x 79 (value :  0.7231) is classified as Class2
Point x 80 (value :  0.3543) is classified as Class1
Point x 81 (value :  0.7346) is classified as Class2
Point x 82 (value :  0.4147) is classified as Class1
Point x 83 (value :  0.3320) is classified as Class1
Point x 84 (value :  0.0376) is classified as Class1
Point x 85 (value :  0.6599) is classified as Class2
Point x 86 (value :  0.8521) is classified as Class2
Point x 87 (value :  0.7628) is classified as Class2
Point x 88 (value :  0.7053) is classified as Class2
Point x 89 (value :  0.2535) is classified as Class1
Point x 90 (value :  0.5581) is classified as Class2
Point x 91 (value :  0.5090) is classified as Class2
Point x 92 (value :  0.1859) is classified as Class1
Point x 93 (value :  0.1852) is classified as Class1
```

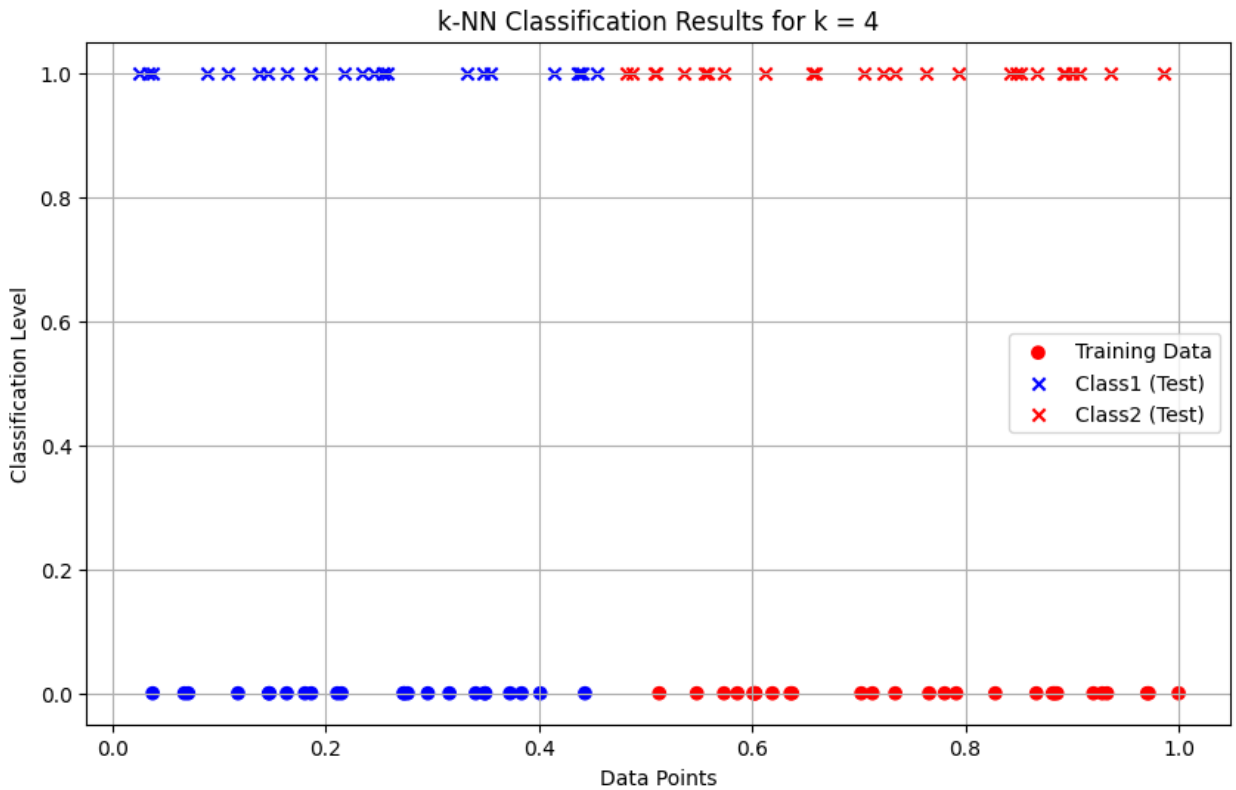
```
Point x 94 (value :  0.0245) is classified as Class1
Point x 95 (value :  0.1084) is classified as Class1
Point x 96 (value :  0.6561) is classified as Class2
Point x 97 (value :  0.9366) is classified as Class2
Point x 98 (value :  0.1635) is classified as Class1
Point x 99 (value :  0.8427) is classified as Class2
Point x 100 (value :  0.4878) is classified as Class2

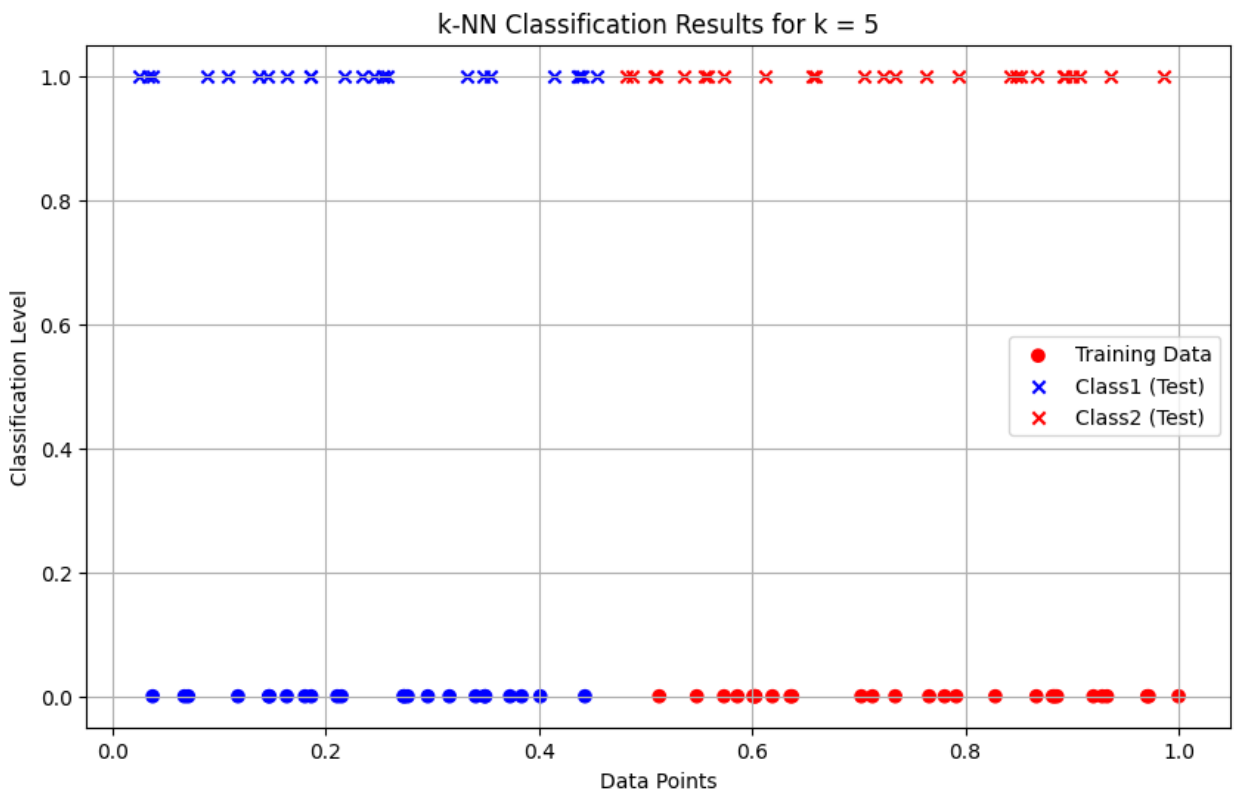

Classification complete.
```

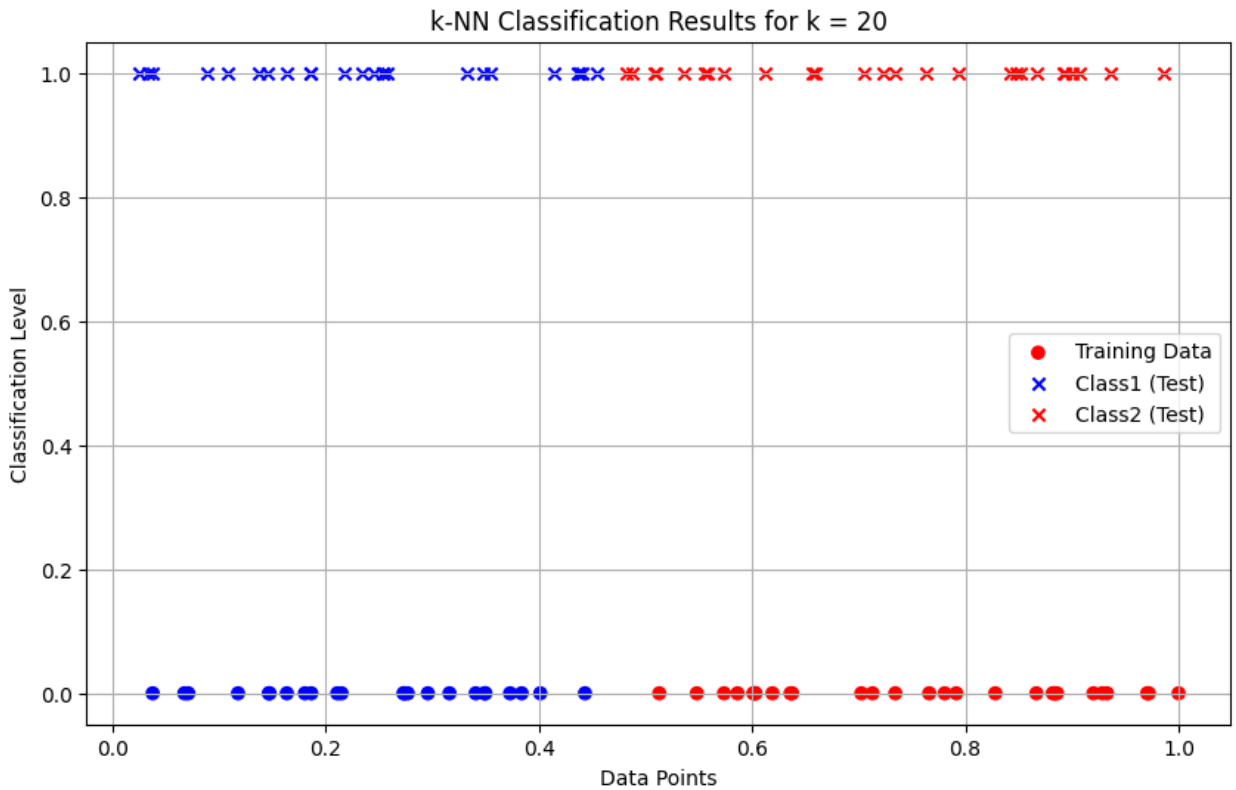


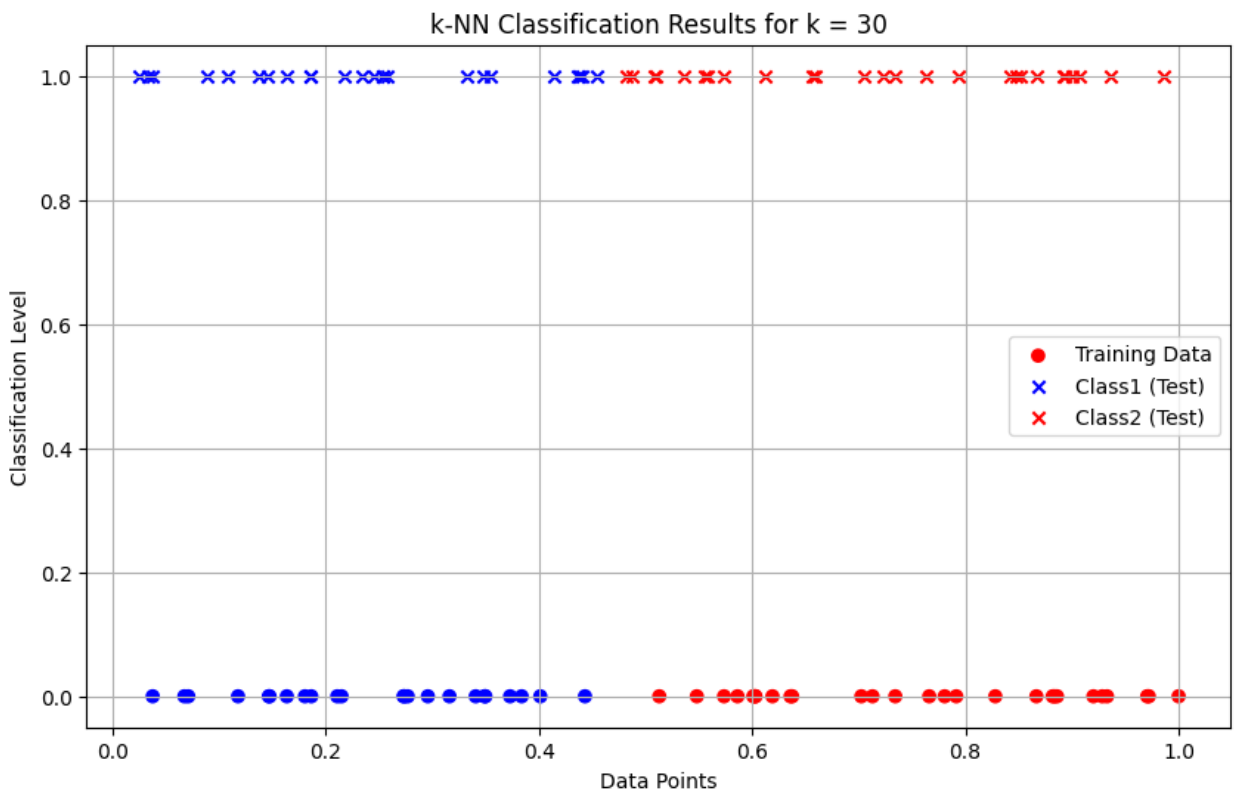

k-NN Classification Results for k = 2
Classification Level
Data Points
Training Data
Class1 (Test)
Class2 (Test)
0.0
0.2
0.4
0.6
0.8
1.0


k-NN Classification Results for k = 3
Classification Level
Data Points
Training Data
Class1 (Test)
Class2 (Test)
0.0
0.2
0.4
0.6
0.8
1.0


k-NN Classification Results for k = 4
Classification Level
Data Points
Training Data
Class1 (Test)
Class2 (Test)
0.0
0.2
0.4
0.6
0.8
1.0


k-NN Classification Results for k = 5
Classification Level
Data Points
Training Data
Class1 (Test)
Class2 (Test)
0.0
0.2
0.4
0.6
0.8
1.0


k-NN Classification Results for k = 20
Classification Level
Data Points
Training Data
Class1 (Test)
Class2 (Test)
0.0
0.2
0.4
0.6
0.8
1.0


k-NN Classification Results for k = 30
Classification Level
Data Points
Training Data
Class1 (Test)
Class2 (Test)
0.0
0.2
0.4
0.6
0.8
1.0

6. Implement the non-parametric Locally Weighted Regression algorithm in order to fit data points. Select appropriate data set for your experiment and draw graphs.

```
import numpy as np
import matplotlib.pyplot as plt

def gaussian_kernel (x, xi, tau) :
    return np.exp (-np.sum ((x - xi) ** 2) / (2 * tau ** 2))

def locally_weighted_regression (x, X, y, tau) :
    m = X.shape [0]
    weights = np.array ([gaussian_kernel (x, X [i], tau) for i in
range (m)])
    W = np.diag (weights)

    X_transpose_W = X.T @ W
    theta = np.linalg.inv (X_transpose_W @ X) @ X_transpose_W @ y

    return x @ theta

np.random.seed (42)

X = np.linspace (0, 2 * np.pi, 100)
y = np.sin (X) + 0.1 * np.random.randn (100)

X_bias = np.c_ [np.ones (X.shape), X]
x_test = np.linspace (0, 2 * np.pi, 200)
x_test_bias = np.c_ [np.ones (x_test.shape), x_test]

tau = 0.5

y_pred = np.array ([locally_weighted_regression (xi, X_bias, y, tau)
for xi in x_test_bias])

plt.figure (figsize = (10, 6))
plt.scatter (X, y, color = 'red', label = 'Training Data', alpha =
0.7)

plt.plot (x_test, y_pred, color = 'blue', label = f'LWR Fit (tau =
{tau})', linewidth = 2)

plt.xlabel ('X - axis ------>', fontsize = 12)
plt.ylabel ('Y - axis ------>', fontsize = 12)

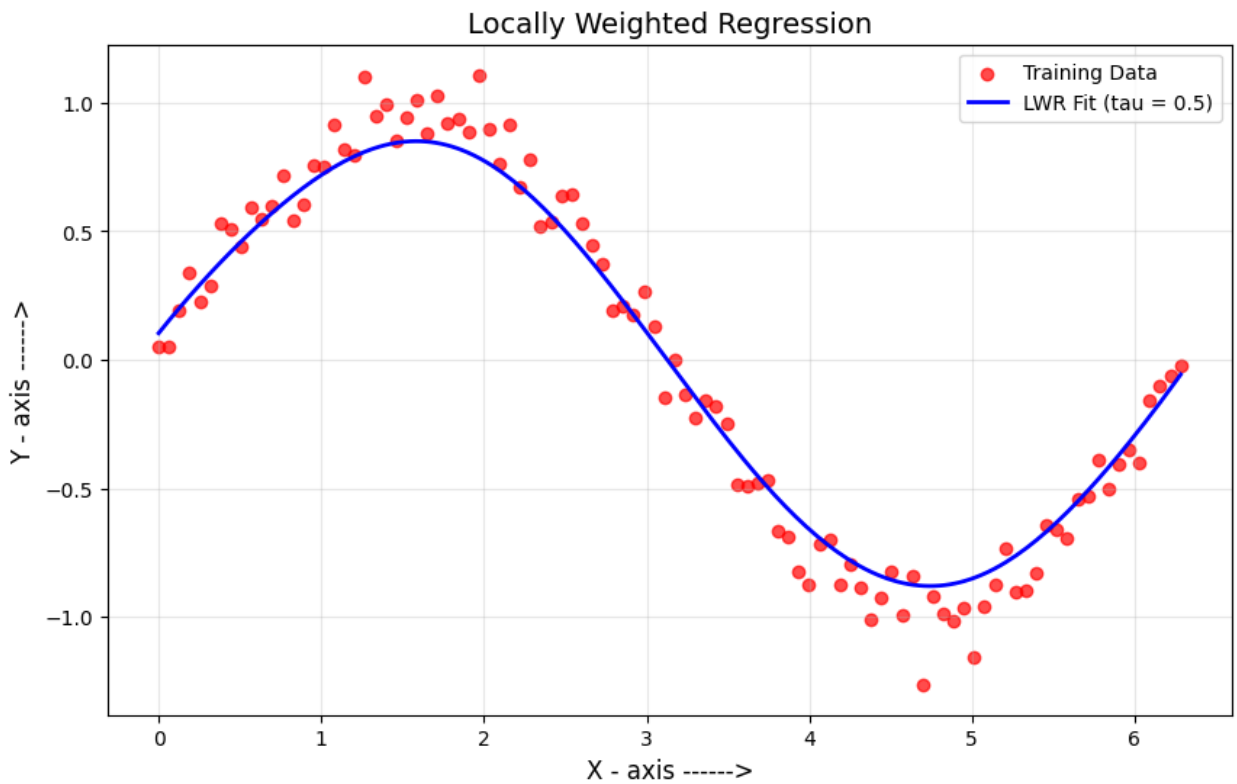
plt.title ('Locally Weighted Regression', fontsize = 14)
```

```
plt.legend (fontsize = 10)
plt.grid (alpha = 0.3)

plt.savefig ('locally_weighted_regression.png', dpi = 300, bbox_inches
= 'tight')
print ("LWR saved as 'locally_weighted_regression.png'")

plt.show ()
```

```
LWR saved as 'locally_weighted_regression.png'
```



7. Develop a program to demonstrate the working of Linear Regression and Polynomial Regression. Use Boston Housing Dataset for Linear Regression and Auto MPG Dataset (for vehicle fuel efficiency prediction) for Polynomial Regression.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures, StandardScaler
from sklearn.pipeline import make_pipeline
from sklearn.metrics import mean_squared_error, r2_score

def linear_regression_california () :
    housing = fetch_california_housing (as_frame = True)

    X = housing.data [["AveRooms"]]
    y = housing.target

    X_train, X_test, y_train, y_test = train_test_split (X, y,
                                            test_size = 0.2,
                                            random_state = 42)

    model = LinearRegression ()
    model.fit (X_train, y_train)

    y_pred = model.predict (X_test)

    plt.scatter (X_test, y_test, color = "blue", label = "Actual")
    plt.plot (X_test, y_pred, color = "red", label = "Predicted")

    plt.xlabel ("Average number of rooms (AveRooms)")
    plt.ylabel ("Median value of homes ($100,000)")

    plt.title ("Linear Regression - California Housing Dataset")
    plt.legend ()

    plt.savefig ('linear_regression.png', dpi = 300, bbox_inches =
'tight')
```

```
    plt.show ()

    print ("Linear Regression - California Housing Dataset")
    print ("Mean Squared Error:", mean_squared_error (y_test, y_pred))
    print ("R^2 Score:", r2_score (y_test, y_pred))

    print ("Linear Regression saved as 'linear_regression.png'")

def polynomial_regression_auto_mpg () :
    url =
"https://archive.ics.uci.edu/ml/machine-learning-databases/auto-mpg/
auto-mpg.data"

    column_names = ["mpg", "cylinders", "displacement", "horsepower",
"weight", "acceleration", "model_year", "origin"]
    data = pd.read_csv (url, sep = '\s+', names = column_names,
na_values = "?")
    data = data.dropna ()

    X = data ["displacement"].values.reshape (-1, 1)
    y = data ["mpg"].values

    X_train, X_test, y_train, y_test = train_test_split (X, y,
                                            test_size = 0.2,
                                            random_state = 42)

    poly_model = make_pipeline (PolynomialFeatures (degree = 2),
StandardScaler (), LinearRegression ())
    poly_model.fit (X_train, y_train)

    y_pred = poly_model.predict (X_test)

    plt.scatter (X_test, y_test, color = "blue", label = "Actual")
    plt.scatter (X_test, y_pred, color = "red", label = "Predicted")

    plt.xlabel ("Displacement")
    plt.ylabel ("Miles per gallon (mpg)")

    plt.title ("Polynomial Regression - Auto MPG Dataset")
    plt.legend ()

    plt.savefig ('polynomial_regression.png', dpi = 300, bbox_inches =
'tight')

    plt.show()

    print ("Polynomial Regression - Auto MPG Dataset")
    print ("Mean Squared Error:", mean_squared_error (y_test, y_pred))
    print ("R^2 Score:", r2_score (y_test, y_pred))
```

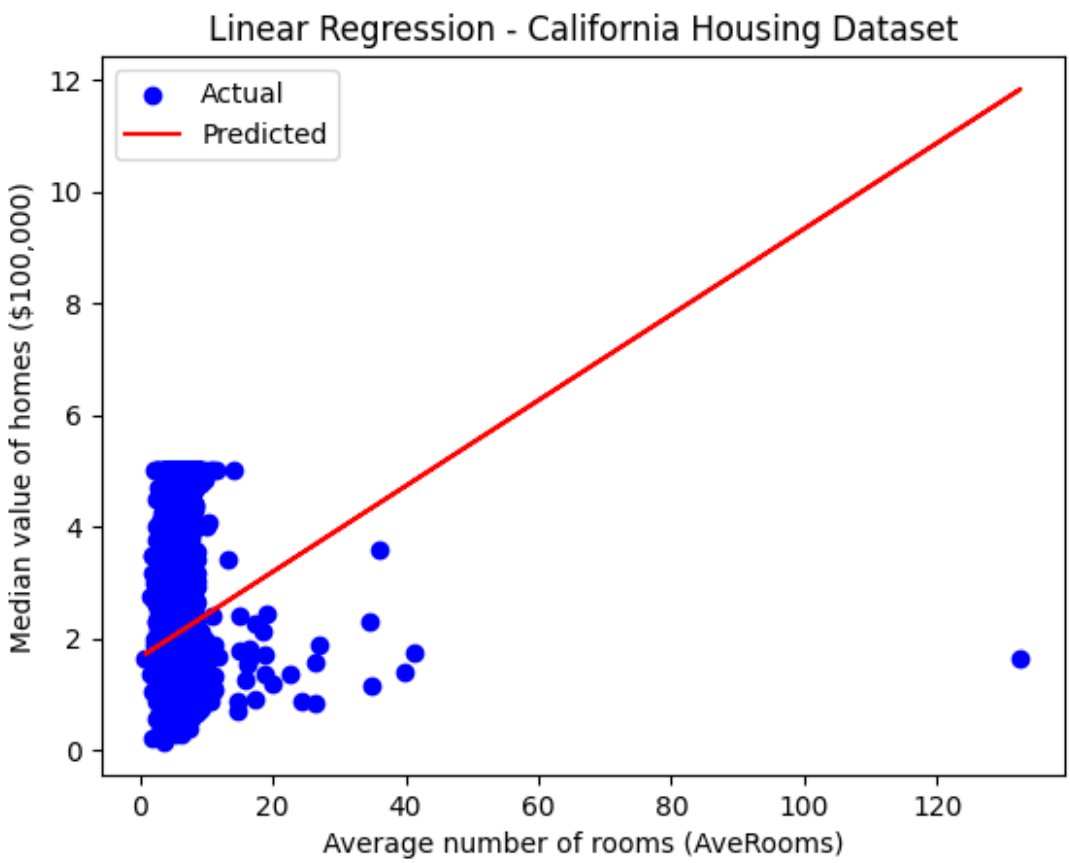
```
    print ("Polynomial Regression saved as
'Polynomial_regression.png'")

if __name__ == "__main__" :
    print ("Demonstrating Linear Regression and Polynomial Regression\
n")

    linear_regression_california ()
    polynomial_regression_auto_mpg ()
```

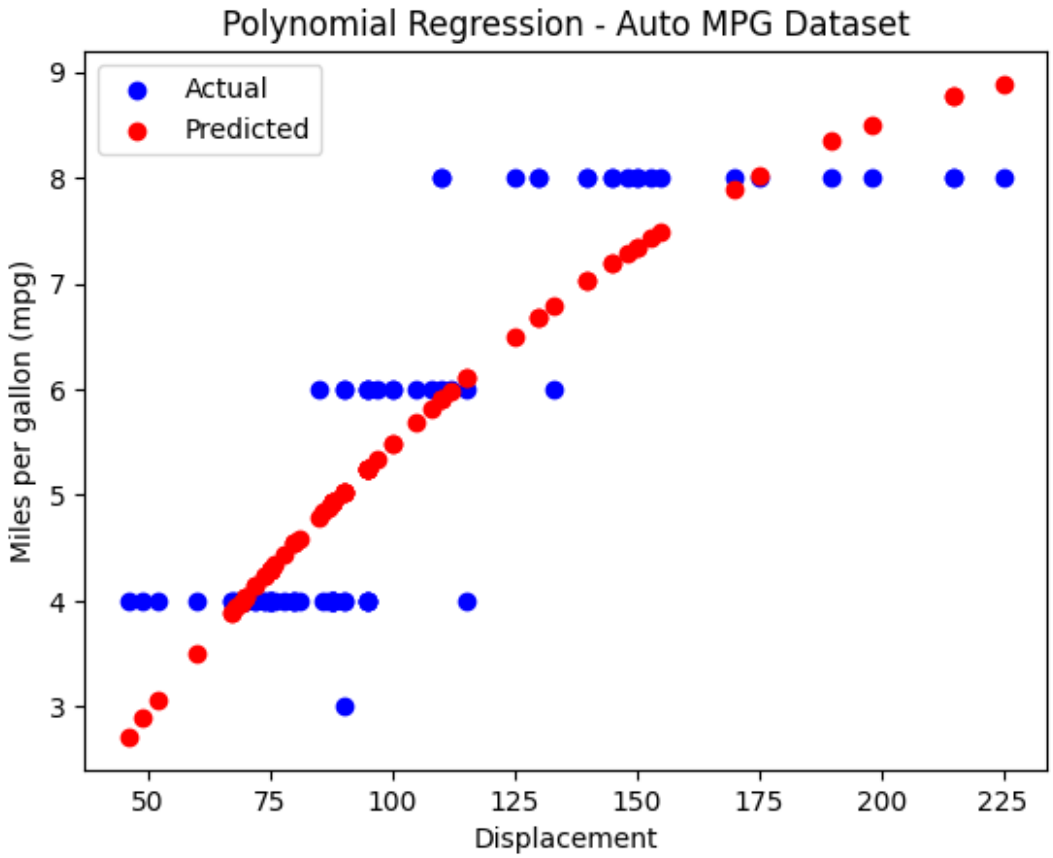
```
<>:50: SyntaxWarning: invalid escape sequence '\s'
<>:50: SyntaxWarning: invalid escape sequence '\s'
C:\Users\HP\AppData\Local\Temp\ipykernel_19216\3216481499.py:50:
SyntaxWarning: invalid escape sequence '\s'
  data = pd.read_csv (url, sep = '\s+', names = column_names,
na_values = "?")

Demonstrating Linear Regression and Polynomial Regression
```




```
Linear Regression - California Housing Dataset
Mean Squared Error: 1.2923314440807299
```

```
R^2 Score: 0.013795337532284901
Linear Regression saved as 'linear_regression.png'
```



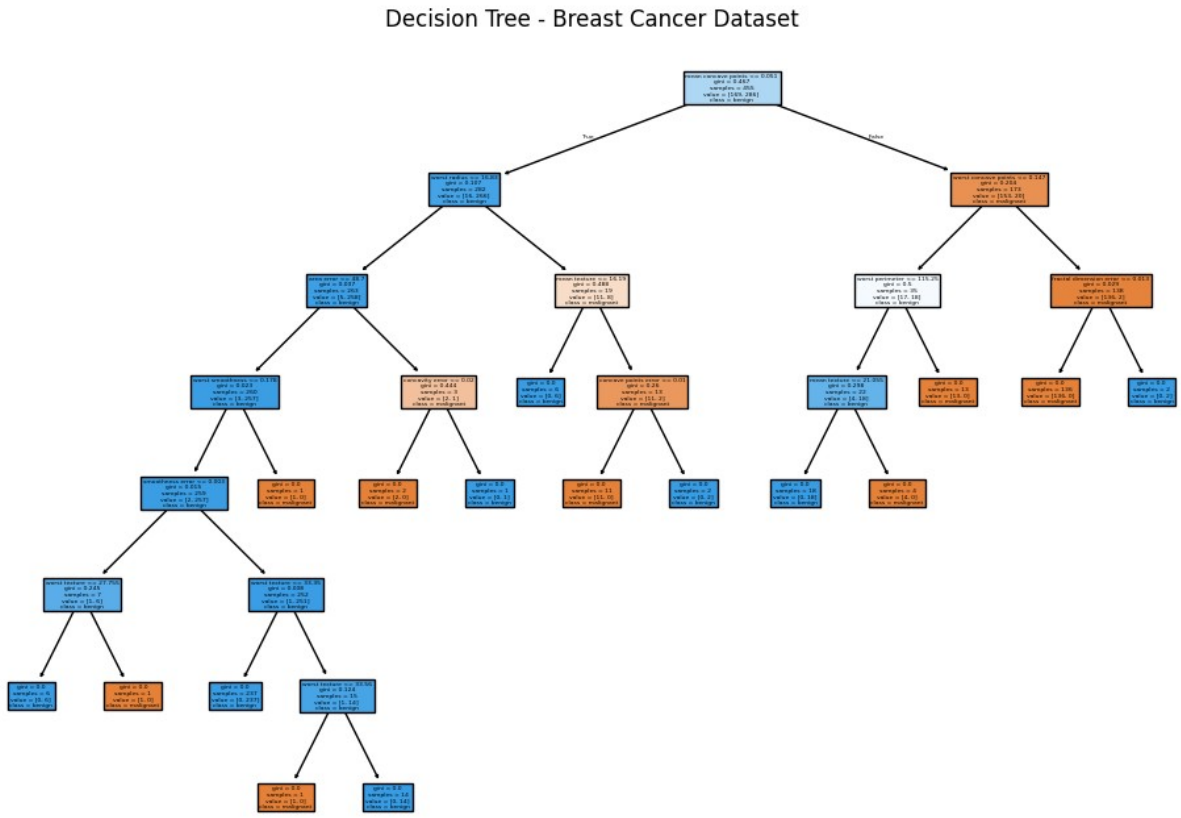

```
Polynomial Regression - Auto MPG Dataset
Mean Squared Error: 0.743149055720586
R^2 Score: 0.7505650609469626
Polynomial Regression saved as 'Polynomial_regression.png'
```

8. Develop a program to demonstrate the working of the decision tree algorithm. Use Breast Cancer Data set for building the decision tree and apply this knowledge to classify a new sample.

```
# Importing necessary libraries
import numpy as np
```

```python
import matplotlib.pyplot as plt

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score
from sklearn import tree

data = load_breast_cancer()

X = data.data
y = data.target

X_train, X_test, y_train, y_test = train_test_split (X, y,
                                                      test_size = 0.2,
                                                      random_state =
42)

clf = DecisionTreeClassifier (random_state = 42)

clf.fit (X_train, y_train)
y_pred = clf.predict (X_test)

accuracy = accuracy_score (y_test, y_pred)
print (f"Model Accuracy: {accuracy * 100: .2f}%")

new_sample = np.array ([X_test[0]])
prediction = clf.predict (new_sample)

prediction_class = "Benign" if prediction == 1 else "Malignant"

print (f"Predicted Class for the new sample: {prediction_class}\n")

plt.figure (figsize = (12, 8))
tree.plot_tree (clf, filled = True,
                feature_names = data.feature_names,
                class_names = data.target_names)

plt.title ("Decision Tree - Breast Cancer Dataset")

plt.savefig ('decision_tree.png', dpi = 300, bbox_inches = 'tight')

plt.show ()

print ("\nDecision Tree - Breast Cancer Dataset saved as
'decision_tree.png'\n")
```

```
Model Accuracy:  94.74%
Predicted Class for the new sample: Benign
```




```
Decision Tree - Breast Cancer Dataset saved as 'decision_tree.png'
```

9. Develop a program to implement the Naive Bayesian classifier considering Olivetti Face Data set for training. Compute the accuracy of the classifier, considering a few test data sets.

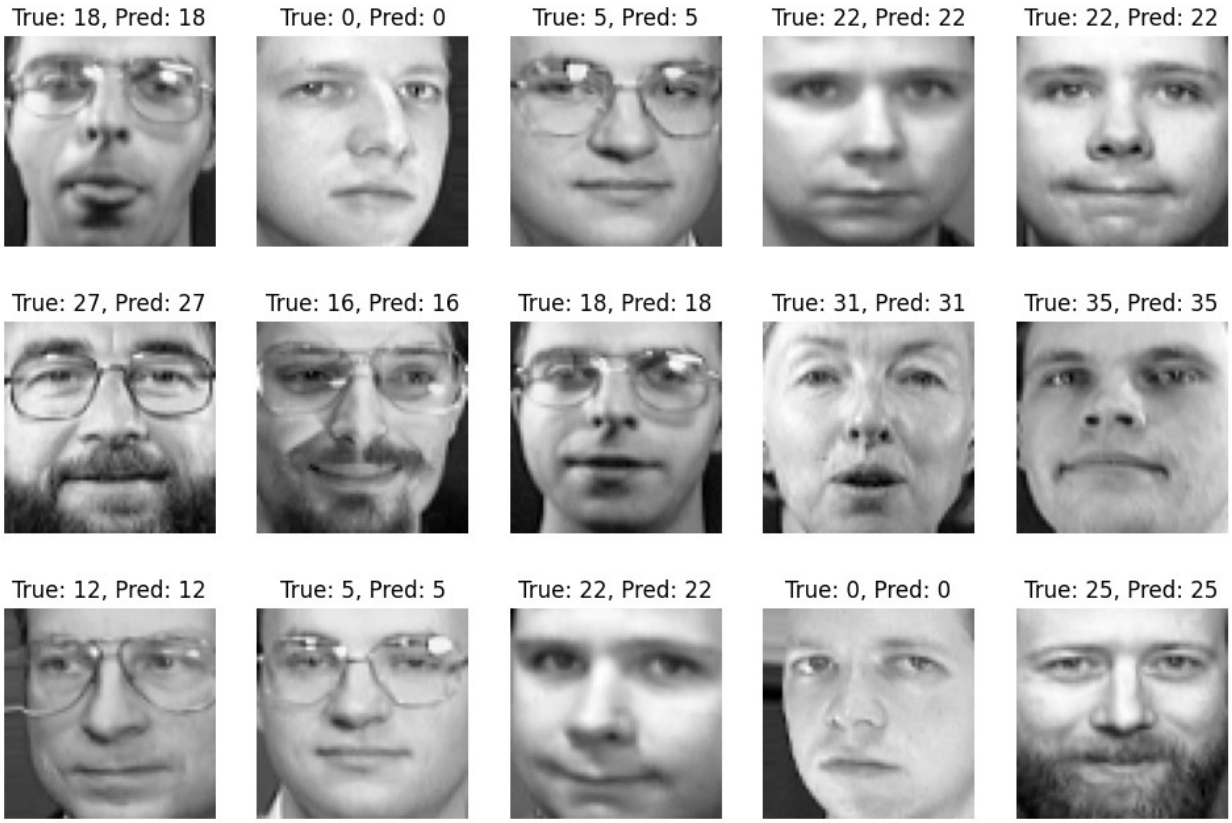
```
import numpy as np
import matplotlib.pyplot as plt

from sklearn.datasets import fetch_olivetti_faces
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, classification_report,
confusion_matrix

data = fetch_olivetti_faces (shuffle = True, random_state = 42)
```

```
X = data.data
y = data.target

X_train, X_test, y_train, y_test = train_test_split (X, y,
                                                     test_size = 0.3,
                                                     random_state =
42)

gnb = GaussianNB ()
gnb.fit (X_train, y_train)

y_pred = gnb.predict (X_test)

accuracy = accuracy_score (y_test, y_pred)

print (f'Accuracy: {accuracy * 100: .2f}%')

print ("\nClassification Report:")
print (classification_report (y_test, y_pred, zero_division = 1))

print ("\nConfusion Matrix:")
print (confusion_matrix (y_test, y_pred))

cross_val_accuracy = cross_val_score (gnb, X, y, cv = 5, scoring =
'accuracy')
print (f'\nCross-validation accuracy: {cross_val_accuracy.mean () *
100: .2f}%\n')

fig, axes = plt.subplots (3, 5, figsize = (12, 8))

for ax, image, label, prediction in zip (axes.ravel (), X_test,
y_test, y_pred) :
    ax.imshow (image.reshape (64, 64), cmap = plt.cm.gray)
    ax.set_title (f"True: {label}, Pred: {prediction}")

    ax.axis ('off')

plt.savefig ('nb_classifier.png', dpi = 300, bbox_inches = 'tight')

plt.show ()

print ("\nOlivetti Face Classification is saved as
'nb_classifier.png'\n")
```

```
Accuracy:  80.83%

Classification Report:
              precision    recall  f1-score   support

           0       0.67      1.00      0.80         2
```

```
           1       1.00      1.00      1.00         2
           2       0.33      0.67      0.44         3
           3       1.00      0.00      0.00         5
           4       1.00      0.50      0.67         4
           5       1.00      1.00      1.00         2
           7       1.00      0.75      0.86         4
           8       1.00      0.67      0.80         3
           9       1.00      0.75      0.86         4
          10       1.00      1.00      1.00         3
          11       1.00      1.00      1.00         1
          12       0.40      1.00      0.57         4
          13       1.00      0.80      0.89         5
          14       1.00      0.40      0.57         5
          15       0.67      1.00      0.80         2
          16       1.00      0.67      0.80         3
          17       1.00      1.00      1.00         3
          18       1.00      1.00      1.00         3
          19       0.67      1.00      0.80         2
          20       1.00      1.00      1.00         3
          21       1.00      0.67      0.80         3
          22       1.00      0.60      0.75         5
          23       1.00      0.75      0.86         4
          24       1.00      1.00      1.00         3
          25       1.00      0.75      0.86         4
          26       1.00      1.00      1.00         2
          27       1.00      1.00      1.00         5
          28       0.50      1.00      0.67         2
          29       1.00      1.00      1.00         2
          30       1.00      1.00      1.00         2
          31       1.00      0.75      0.86         4
          32       1.00      1.00      1.00         2
          34       0.25      1.00      0.40         1
          35       1.00      1.00      1.00         5
          36       1.00      1.00      1.00         3
          37       1.00      1.00      1.00         1
          38       1.00      0.75      0.86         4
          39       0.50      1.00      0.67         5

    accuracy                           0.81       120
   macro avg       0.89      0.85      0.83       120
weighted avg       0.91      0.81      0.81       120


Confusion Matrix:
[[2 0 0 ... 0 0 0]
 [0 2 0 ... 0 0 0]
 [0 0 2 ... 0 0 1]
 ...
 [0 0 0 ... 1 0 0]
 [0 0 0 ... 0 3 0]
```

```
 [0 0 0 ... 0 0 5]]

Cross-validation accuracy:  87.25%
```




```
Olivetti Face Classification is saved as 'nb_classifier.png'
```

10. Develop a program to implement k-means clustering using Wisconsin Breast Cancer data set and visualize the clustering result.

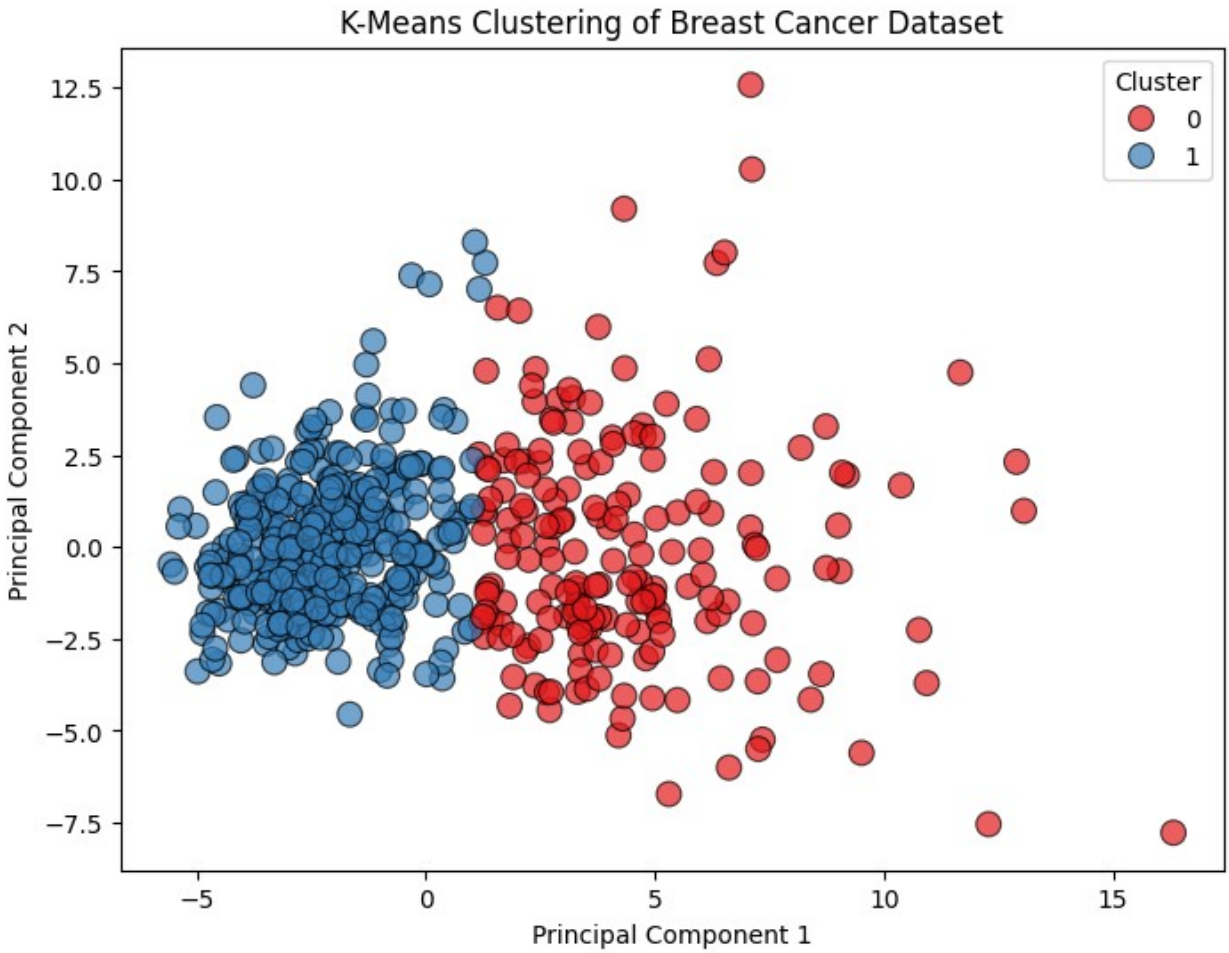
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets         import load_breast_cancer
from sklearn.cluster          import KMeans
```

```
from sklearn.preprocessing    import StandardScaler
from sklearn.decomposition    import PCA
from sklearn.metrics          import confusion_matrix,
classification_report

data = load_breast_cancer ()

X = data.data
y = data.target

scaler = StandardScaler ()
X_scaled = scaler.fit_transform (X)

kmeans = KMeans (n_clusters = 2, random_state = 42)
y_kmeans = kmeans.fit_predict (X_scaled)

print ("Confusion Matrix: ")
print (confusion_matrix (y, y_kmeans))

print ("\nClassification Report: ")
print (classification_report (y, y_kmeans))

pca = PCA (n_components = 2)
X_pca = pca.fit_transform (X_scaled)

df = pd.DataFrame (X_pca, columns = ['PC1', 'PC2'])

df ['Cluster'] = y_kmeans
df ['True Label'] = y

print ('\n\n---- Images are shown below ----\n\n')
#
plt.figure (figsize = (8, 6))

sns.scatterplot (data = df,
                 x = 'PC1', y = 'PC2',
                 hue = 'Cluster', palette = 'Set1', edgecolor =
'black',
                 s = 100, alpha = 0.7)

plt.title ('K-Means Clustering of Breast Cancer Dataset')

plt.xlabel ('Principal Component 1')
plt.ylabel ('Principal Component 2')

plt.legend (title = "Cluster")

plt.savefig ('kmeans_cluster.png', dpi = 300, bbox_inches = 'tight')

plt.show ()
```

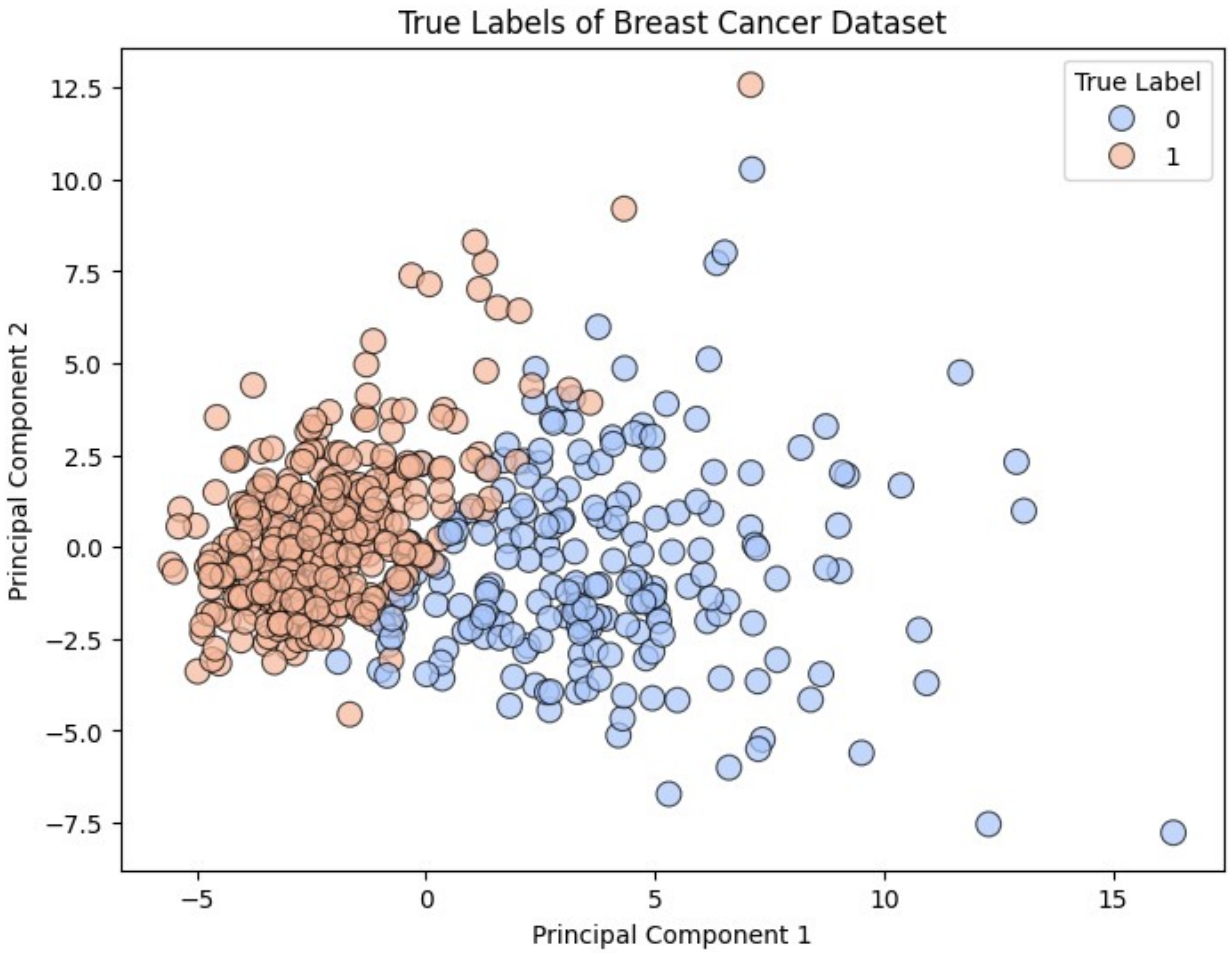
```
print ("K-Means Clustering of Breast Cancer Dataset saved as
'kmeans_cluster.png'\n- - - - - - - -")

#
plt.figure (figsize = (8, 6))

sns.scatterplot (data = df,
                 x = 'PC1', y = 'PC2',
                 hue = 'True Label', palette = 'coolwarm', edgecolor =
'black',
                 s = 100,  alpha = 0.7)

plt.title ('True Labels of Breast Cancer Dataset')

plt.xlabel ('Principal Component 1')
plt.ylabel ('Principal Component 2')

plt.legend (title = "True Label")

plt.savefig ('true_labels.png', dpi = 300, bbox_inches = 'tight')

plt.show ()

print ("True Labels of Breast Cancer Dataset saved as
'true_labels.png'\n- - - - - - - -")

#
plt.figure (figsize = (8, 6))

sns.scatterplot (data = df,
                 x = 'PC1', y = 'PC2',
                 hue = 'Cluster', palette = 'Set1', edgecolor =
'black',
                 s = 100,  alpha = 0.7)

centers = pca.transform (kmeans.cluster_centers_)

plt.scatter (centers [:, 0], centers [:, 1],
             s = 200,
             c = 'red', marker = 'X', label = 'Centroids')

plt.title ('K-Means Clustering with Centroids')

plt.xlabel ('Principal Component 1')
plt.ylabel ('Principal Component 2')

plt.legend (title = "Cluster")

plt.savefig ('kmeans_centroids.png', dpi = 300, bbox_inches = 'tight')

plt.show()
```

```
print ("K-Means Clustering with Centroids saved as
'kmeans_centroids.png'\n\n---- End of Execution ----\n")
```

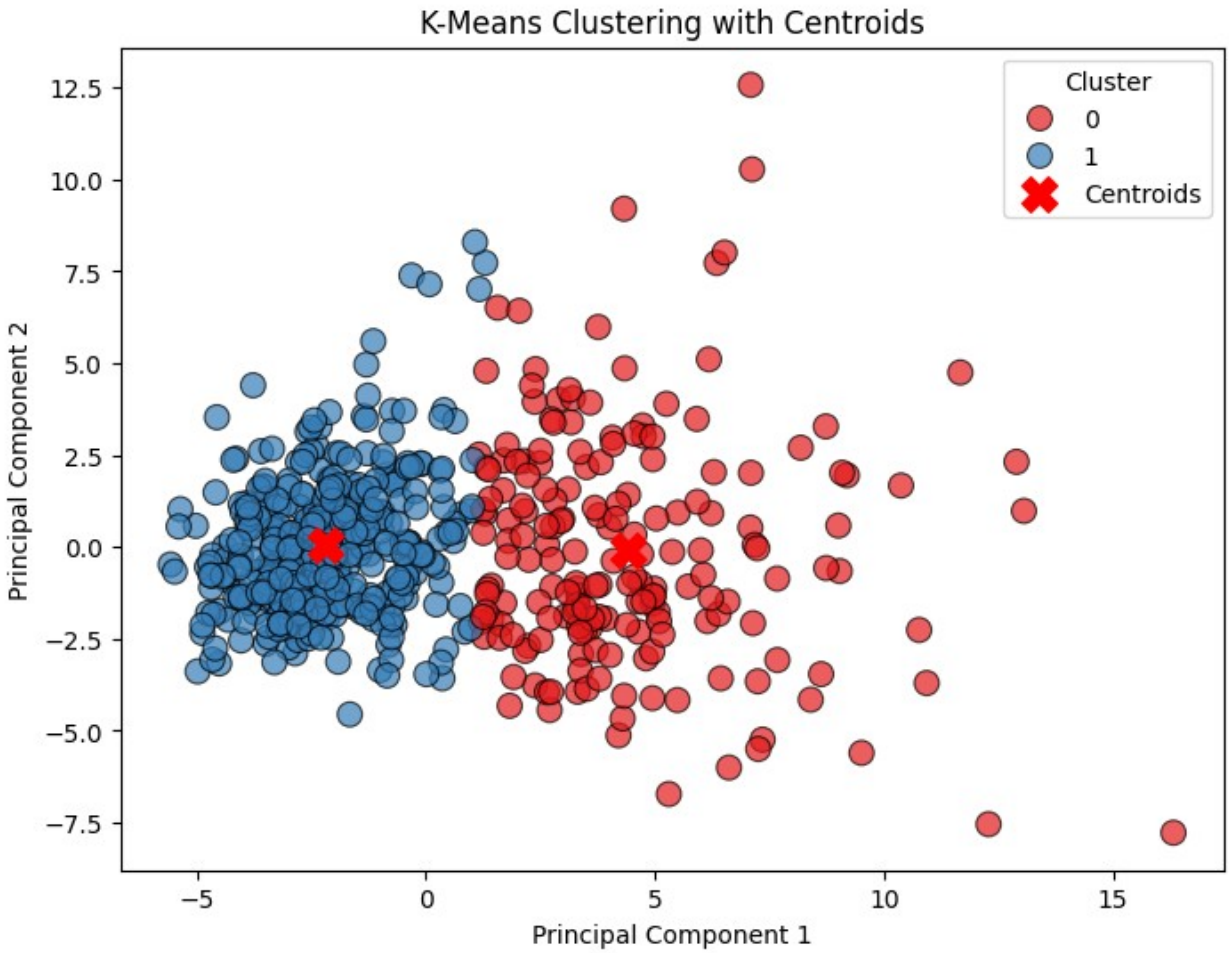
```
Confusion Matrix:
[[175  37]
 [ 13 344]]

Classification Report:
              precision    recall  f1-score   support

           0       0.93      0.83      0.88       212
           1       0.90      0.96      0.93       357

    accuracy                           0.91       569
   macro avg       0.92      0.89      0.90       569
weighted avg       0.91      0.91      0.91       569



---- Images are shown below ----
```




```
K-Means Clustering of Breast Cancer Dataset saved as
'kmeans_cluster.png'
- - - - - - - -
```




```
True Labels of Breast Cancer Dataset saved as 'true_labels.png'
- - - - - - - -
```




```
K-Means Clustering with Centroids saved as 'kmeans_centroids.png'

---- End of Execution ----
```